

Exceptive-additive *besides*: a unified semantic treatment*

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Abstract

This paper investigates English exceptive-additive constructions introduced by *besides*. Like the better-studied pure exceptives introduced by *but*, *besides* contributes an inference of exception when it occurs with quantifiers that exhibit universal force (e.g., *every* or *no*). However, while exceptives are incompatible with quantifiers that have existential force, this is not the case for *besides*. In such contexts, *besides* contributes an inference of addition. We propose a unified compositional account that explains the exceptive-additive behavior of *besides* and its distinctions from exceptives like *but*.

Our analysis extends the treatment of exceptives framed in terms of *EXH* (Gajewski 2013, Hirsch 2016, Crnič 2021, 2024) to exceptive-additives. We propose that, while both *besides* and *but* share a core contribution of domain subtraction, the methods for deriving alternatives differ between the two types of constructions. For quantificational claims with *but*, the alternatives are constructed by substituting the DP following *but* with other DPs. In contrast, the only alternative for a quantificational claim with *besides* that is distinct from the original claim has no subtraction at all.

By reducing the set of alternatives, our account explains the compatibility of *besides* with *exactly-n* numerals and captures the additive inference observed in these cases. At the same time, it still accounts for the exceptive-like behavior of *besides* when paired with universal and negative quantifiers. To generalize this approach to all modified numerals, we adopt a decompositional theory of their semantics. According to this theory, modified numerals are treated as quantifiers over degrees that can optionally move at LF, leaving behind a constituent equivalent in meaning to an *exactly-n* claim. We propose that the additive inference is computed under the scope of a numeral on this constituent. We suggest that a similar approach to the treatment of existential *some* can account for its interaction with *besides*. Finally, we extend our analysis to wh-questions, exploring the implications of applying *EXH* to weak answers.

1 Introduction: exceptive and exceptive-additive constructions

Exceptive constructions, such as those introduced by *except* and *but* in English, have received significant attention and discussion in the literature. In this paper, our focus lies on the semantics of a less explored construction introduced by *besides*. While it shares some core properties with exceptives, it also differs substantially from them.

The two main semantic properties of exceptive constructions are their uniform applicability to both universal and negative quantifiers, contributing an exceptive inference in such contexts (Hoeksema 1987, von Stechow 1993), and their inability to combine with quantifiers of any other quantificational force (Horn 1989). *Besides* behaves like exceptives concerning the first property, but crucially, does not possess similar restrictions on quantifier compatibility.

To illustrate the first property of exceptives, let's consider examples (1) and (2). In (1), where the quantificational generalization is positive, the use of *except* results in a negative exception inference. Conversely, with the negative quantificational generalization in (2), the exception inference becomes positive. Additionally, *except* and *but* contribute the inference that the individual to which the exceptive inference applies belongs to the restrictor set. This particular inference is referred to as *containment* in the exceptive literature. In this case, this is the inference that Ann is a student.

- | | | |
|-----|---|----------------|
| (1) | Every student <i>but/except</i> Ann passed. | |
| | ↗ Every student who is not Ann passed | quantification |
| | ↗ Ann didn't pass | exception |
| | ↗ Ann is a student | containment |
| (2) | No student <i>but/except</i> Ann passed. | |
| | ↗ No student who is not Ann passed | quantification |
| | ↗ Ann passed | exception |
| | ↗ Ann is a student | containment |

Similar to exceptives, *besides* can be used with both universal and negative quantifiers, resulting in the exception inference. The containment inference is also present in such contexts. This is exemplified in (3) and (4) below.

- | | | |
|-----|--|----------------|
| (3) | Every student <i>besides</i> Ann passed. | |
| | ↗ Every student who is not Ann passed | quantification |
| | ↗ Ann didn't pass | exception |
| | ↗ Ann is a student | containment |
| (4) | No student <i>besides</i> Ann passed. | |
| | ↗ No student who is not Ann passed | quantification |
| | ↗ Ann passed | exception |
| | ↗ Ann is a student | containment |

The second key semantic property of exceptive constructions, namely the restriction on their co-occurrence with non-universal quantifiers, is illustrated in (5). Examples of this kind were reported to be ungrammatical for native speakers of English (Horn 1989, von Stechow 1993).¹

¹There is debate in the literature regarding the compatibility of exceptive constructions with quantifiers that are close to the endpoint of a Horn scale, such as *many*, *most*, and *few*. Such expressions might form an exception to this generalization.

- (5) a. *Some student *but/except* Ann passed.
- b. *At least one/exactly one student *but/except* Ann passed.
- c. *At most one/more than one student *but/except* Ann passed.

This property sets *besides* apart from exceptives: it is grammatical in such cases, as shown in (6), (7), and (8). More interestingly and importantly, the inference contributed by *besides* in all these cases is positive (Ann passed). In this regard, these cases are similar to the case with *no student* in (4), even though no negative quantifier is present in (6) and (7). The sentence in (6) can be paraphrased as follows: in addition to Ann, some other students passed. Descriptively, we will call these readings additive, and we will categorize *besides* as an exceptive-additive expression, in contrast to pure exceptives like *but* and *except*. The containment inference is present in all cases. In this regard the exceptive and the additive uses of *besides* pattern like exceptives.

- | | | |
|-----|---|----------------|
| (6) | Some student(s) <i>besides</i> Ann passed. | |
| | ↗ Some student(s) who is not Ann passed | quantification |
| | ↗ Ann passed | addition |
| | ↗ Ann is a student | containment |
| (7) | At least/more than/exactly one student <i>besides</i> Ann passed. | |
| | ↗ At least/more than one student who is not Ann passed | quantification |
| | ↗ Ann passed | addition |
| | ↗ Ann is a student | containment |
| (8) | At most/fewer than two students <i>besides</i> Ann passed. | |
| | ↗ At most/fewer than two students who is not Ann passed | quantification |
| | ↗ Ann passed | addition |
| | ↗ Ann is a student | containment |

It is worth pointing out that *some*, *at least/more than one* are upward monotone on their restrictor and scope, *exactly one* is non-monotone on both its arguments, and *at most/less than one* are downward monotone on both of their arguments. However, these differences in monotonicity seem to be irrelevant for the inferences licensed by the addition of *besides*. Given the relevance of monotonicity for pure exceptives, this fact poses a challenge for a unified semantic treatment of exceptive and exceptive-additive cases.

Another context in which the behavior of *besides* differs from that of exceptives is in *which*-questions. While *except* and *but* are ungrammatical in *which*-questions, as demonstrated in (9) (von Fintel 1994), *besides* is grammatical in these contexts and, once again, yields the additive inference (Vostrikova 2019). The containment inference is present here, as in all cases considered above.

-
- (i) a. ?Most students *but/except* Ann passed.
 - b. ?Many students *but/except* Ann passed.
 - c. ?Few students *but/except* Ann passed.

von Fintel (1994), Moltmann (1995) and Hirsch (2016), on the one hand, consider these cases to be ungrammatical and develop theories to explain this ungrammaticality. von Fintel (1994) discusses the compatibility of exceptives introduced by *except for* with such quantifiers. On the other hand, García Álvarez (2008) argues that exceptives introduced by *but* and *except* are grammatical in such contexts, a position also supported by Peters and Westersthål (2006), Peters and Westersthål (2023) and Nadathur and Lassiter (2018). Peters and Westersthål (2023) presents compelling examples where quantifiers like *most* and *few* are modified by various types of exceptives, including those introduced by *but*. We leave this issue for future research.

- (9) a. *Which student(s) *but/except* Ann passed?
 b. Which students *besides* Ann passed?
 $\rightsquigarrow$ Which student(s) who is not Ann passed? question
 $\rightsquigarrow$ Ann passed addition
 $\rightsquigarrow$ Ann is a student containment

The generalization emerging from the review of this empirical picture is given in (10).

- (10) **The exceptive-additive generalization:** Exceptive-additive *besides* behaves in parallel to exceptives in contexts where exceptives are acceptable. In contexts where exceptives are ungrammatical, *besides* is grammatical and obtains the additive reading.

In this work, our objective is to provide a semantic theory that accurately captures the exceptive-like behavior of *besides* with universal and negative quantifiers, as well as its additive behavior with existentials, modified numerals, and *wh*-questions. Additionally, we aim to provide an explanation for the differences between *besides* and exceptives such as *except* and *but*.

There is consensus in the literature that the explanation of the restrictions on the use of pure exceptives should be semantic. As will be seen the reason for this is that the empirical pattern is predictable from the logical properties of the quantifiers involved. Given the first part of the generalization in (10), this explanation should carry over to exceptive-additive *besides*. Adopting such an account wholesale, however, would also have the consequence that *besides* with non-universal quantifiers should come out as ungrammatical. Notice incidentally that the fact that *besides* always gives rise to the additive interpretation in these contexts again suggests that one is dealing with a systematic and predictable pattern.

Given the parallel to pure exceptives when occurring with universal quantifiers and the systematic appearance of the additive interpretation with all other quantifiers, we advocate a ‘no ambiguity’ account and propose a unified treatment for both the exceptive and additive uses of *besides*. The core idea behind our proposal is that the meaning of a sentence containing *besides* is derived using the same mechanisms employed in the semantics of exceptives: set subtraction and EXH (von Stechow 1993, Gajewski 2013, Hirsch 2016, Crnić 2021). The distinction between these two types of constructions lies in how the alternatives are constructed: *besides* uses a smaller set of alternatives. In this case, EXH asserts the quantificational claim with subtraction made by *besides* and negates the same claim without the subtraction. We demonstrate that this approach naturally yields a positive inference if the quantifier is negative or non-monotone (such as *exactly*) and a negative inference if the quantifier is universal. Additionally, we extend this treatment to all cases of modified numerals by assuming that they are quantificational expressions that can move, leaving behind a constituent with a meaning equivalent to an *exactly n* claim. By placing EXH above this constituent but below the scope of the numeral, we naturally account for the additive inference observed in all cases of modified numerals regardless of the properties of the numeral itself. This treatment is further extended to existential constructions and *wh*-questions. Regarding the latter we also propose a natural generalization of EXH to non-propositional expressions. To the best of our knowledge, this is the first proposal that comprehensively accounts for all these different uses of *besides*.

It is worth noting that constructions with similar exceptive-additive properties have been reported to exist in many other languages (Mayer (1993) discusses such a

construction for German, and Sevi (2008) does so for Hebrew. Constructions that can switch between the exceptive and additive meanings are reported to exist in Yiddish, Hebrew, Estonian, Ukrainian, Lithuanian, Nanai, Kalmyk, and Ossetian in (Oskolskaya 2011). Vostrikova (2019) describes exceptive-additive constructions in Russian, Turkish, Hindi, Bulgarian, Persian. Liu (2019) discusses such a construction in Mandarin.) However, this work will solely focus on English *besides*, leaving open the possibility of extending the proposed account to other languages.

The discussion will be structured as follows: Section 2 will provide a formal background on contemporary semantic theories of exceptive constructions, which will serve as our starting point, and will also review existing theories on *besides*. In Section 3, we will present our proposal for the semantics of *besides* and explain its interaction with universal quantifiers and numerals modified by *exactly*. Section 4 extends the approach to upward-entailing and downward-entailing numerals. Section 5 provides arguments for decomposing the meaning contribution of *besides*, drawing parallels to *but*. Section 6 addresses challenges and extensions to the proposed account. First, we observe that *besides* is incompatible with modified numerals in the context of collective predicates. We propose that a minor modification to the theory can account for the contrast between collective and distributive predicates. Specifically, we show that appealing to maximality based on informativity effectively captures this distinction. Next, we discuss challenges posed by the interaction between *besides* and upward-scale predicates. Section 7 extends the account to wh-questions. Finally, Section 8 provides concluding remarks.

2 Background on the semantics of exceptives

Our account of *besides* builds upon existing approaches to the semantics of exceptives, with necessary modifications to accommodate the differences between the two constructions. In this section, we will review the standard approach to the semantics of exceptives and introduce the key theoretical tools employed by our theory.

2.1 The unified treatment of *but* with negative and universal quantifiers

The core idea of behind the a semantic theory of exceptives, which reflects their close connection to quantifiers, is that they impose a restriction on the domain of a quantifier by subtracting an element from the restrictor set (Hoeksema 1987, von Stechow 1993 a.m.o.). In (11), for example, *but* subtracts *Ann* from the set of students, resulting in an *every*-claim made for a smaller domain.

- (11) Every student but Ann passed.

This alone would be similar to domain restriction. In itself it does not account for the exception inference in examples like (11). A widely accepted approach to capturing this inference involves negating alternative quantificational claims that differ from the original only in terms of what is subtracted from the domain (von Stechow 1993, 1994).

Various implementations of this idea have been proposed, with the most current approach using the *EXH* operator and focus alternatives (Gajewski 2008, Hirsch 2016, Crnić 2021). This theoretical framework is particularly appealing as it leverages a mechanism familiar from the analysis of numerous other empirical phenomena, such as scalar implicatures, NPIs, embedded questions, and plurals.

According to this approach, (11) would get an LF like the one shown in (12).

$$(12) \quad [IP_2 \text{ EXH}_{Alt} [IP_1 [\text{every} [\text{student} [\text{but Ann}_F]]] \text{ passed}]]$$

In this semantic framework, the exceptive itself only contributes the inferences of domain subtraction and containment. The lexical entry of *but* can be represented as in (13a) following Hirsch (2016). It is a function that takes an atomic or plural individual (provided by the DP that follows *but*) and a predicate of individuals (represented by the restrictor of the quantifier) to create a new predicate of individuals (plural or atomic). This new predicate is true of individuals that do not overlap with the individual introduced in the exceptive phrase. $\circ$ stands for the ‘overlap’-relation holding between two individuals if they have a subpart in common, as defined in (13b).² The containment inference is treated as a definedness condition. *But* presupposes that the atomic or plural individual following it is a member of the restrictor set.

$$(13) \quad \begin{array}{ll} \text{a.} & \llbracket but \rrbracket^{g,w} = \lambda x_e. \lambda f_{et} : f(x) = 1. \lambda y_e. f(y) = 1 \ \& \ \neg(x \circ y) \\ \text{b.} & x \circ y \text{ iff } \exists z[z \leq x \ \& \ z \leq y] \end{array}$$

With these assumptions, the resulting restrictor set is as shown in (14).

$$(14) \quad \begin{array}{ll} \text{a.} & \llbracket student \ but \ Ann \rrbracket^{g,w} = \lambda y_e. y \text{ is a student}_w \ \& \ \neg(y \circ Ann) \\ & \llbracket student \ but \ Ann \rrbracket^{g,w} \text{ is defined only if } Ann \text{ is a student}_w \end{array}$$

The prejacent of the *EXH* operator obtains the meaning shown in (15). It represents the quantificational claim with *Ann* subtracted from its domain.

$$(15) \quad \llbracket IP_1 \rrbracket^{g,w} = T \text{ iff } \forall x[x \text{ is a student} \ \& \ \neg(x \circ y) \rightarrow x \text{ passed}_w]$$

The exception inference is contributed by the *EXH* operator, which asserts the prejacent in (15) and negates the relevant alternatives. A crucial element of this theory is the way alternatives are constructed. As seen in (12), the complement of *but* is marked with focus. The idea is that the presence of *but* obligatorily triggers the focus marking on the DP following it, and the resulting alternatives must be utilized by *EXH*. Given this, the alternatives are formed by making a substitution in the prejacent in the position following *but*. Following the structural theory of focus alternatives (Katzir 2007, Fox and Katzir 2011), we adopt the assumption that the alternatives are no more complex than the original sentence. Assuming there are four contextually relevant individuals, the set of structural alternatives is as shown in (16).

$$(16) \quad Alt_{str}(IP_1) \subseteq \left\{ \begin{array}{l} \llbracket [every [student [but Ann_F]]] passed \rrbracket \\ \llbracket [every [student [but Bill_F]]] passed \rrbracket \\ \llbracket [every [student [but Carl_F]]] passed \rrbracket \\ \llbracket [every [student [but Dan_F]]] passed \rrbracket \end{array} \right\}$$

We employ the conventional denotation for *EXH* provided in (17). *EXH* asserts its prejacent while negating innocently excludable alternatives (IE). These are alternatives that can be excluded without introducing a contradiction. To be more precise, these alternatives are found in every maximal set that can be negated consistently alongside

²We assume that the domain of individuals forms a join-semilattice (Link 1983) structured by the part-of relation. Individual x is a part of individual y , $x \leq y$, iff there is an individual z such that the individual sum of x and z , $x \oplus z$, is identical to y . An individual without proper part is called an atom, whereas an individual with proper part is referred to as a plurality. $x \leq_{at} y$ if $x \leq y$ and x is an atom.

the assertion of the prejacent (Fox 2007).

- (17) a. $\llbracket EXH_{ALT} \phi \rrbracket^{g,w} = \llbracket \phi \rrbracket^{g,w} \& \forall q [q \in IE(\lambda w'. \llbracket \phi \rrbracket^{g,w'}, ALT) \rightarrow \neg q(w)]$
 b. $IE(p, ALT) = \cap \{ALT' \subseteq ALT : ALT' \text{ is a maximal subset of } ALT \text{ s.t. } ALT'^\neg \cup \{p\} \text{ is consistent}\}$
 c. $ALT'^\neg = \{\neg p' : p' \in ALT'\}$

ALT in (17) is a set of propositions corresponding to the sentences in the set of structural alternatives in (16). Formally, this can be accomplished by imposing the condition that the set of alternatives utilized by EXH must be a subset of the propositions denoted by the sentences in the set of structural alternatives.³

All alternatives other than the prejacent in this case can be consistently negated. The resulting interpretation for the entire sentence is illustrated in (18): the first conjunct presents the quantificational claim with domain subtraction, while the other conjuncts result from negating the alternatives. Together, the two conjuncts entail that Ann did not pass. Considering that every student who is not Ann passed, along with some student who is not Bill, Carl, or Dan not passing, it follows that Ann must be the student who did not pass.

- (18) $\llbracket (12) \rrbracket^{g,w} = T$ iff $\forall x [x \in \{Bill, Carl, Dan\} \rightarrow x \text{ passed}_w] \&$
 $\neg \forall x [x \in \{Ann, Carl, Dan\} \rightarrow x \text{ passed}_w] \&$
 $\neg \forall x [x \in \{Ann, Bill, Dan\} \rightarrow x \text{ passed}_w] \&$
 $\neg \forall x [x \in \{Ann, Bill, Carl\} \rightarrow x \text{ passed}_w]$

This semantic approach effectively captures the interaction between *but* and *no* in (19a) without the need for additional assumptions, i.e., it explains why *but* can apply to both universal and negative quantifiers in a unified manner, accounting for the first key property of exceptives discussed in the introduction. In particular, the sentence in (19a) gets an LF like the one in (19b), parallel to the one in (12).

- (19) a. No student but Ann passed.
 b. $[IP_2 EXH_{Alt} [IP_1 [no [student [but Ann_F]]]] \text{ passed }]]$

EXH asserts its prejacent and negates all other alternatives, resulting in the truth conditions as shown in (20). Assuming the same set of students, if no student who is not Ann passed, and some student who is not Bill, Carl, or Dan passed, it logically follows that Ann must be the student who passed. Consequently, this approach accurately explains the positive inference present in the sentence in (19a).

- (20) $\llbracket (19b) \rrbracket^{g,w} = T$ iff $\neg \exists x [x \in \{Bill, Carl, Dan\} \& x \text{ passed}_w] \&$
 $\exists x [x \in \{Ann, Carl, Dan\} \& x \text{ passed}_w] \&$
 $\exists x [x \in \{Ann, Bill, Dan\} \& x \text{ passed}_w] \&$
 $\exists x [x \in \{Ann, Bill, Carl\} \& x \text{ passed}_w]$

³More precisely:

- (i) a. $\llbracket EXH_{ALT} \phi \rrbracket^{g,w}$ is defined only if $ALT \subseteq \{\lambda w'. \llbracket \phi' \rrbracket^{g,w'} : \phi' \in ALT_{str}(\phi)\}$
 b. $ALT_{str}(\phi) = \{\phi' : \phi' \text{ is derived from } \phi \text{ by replacing the focus marked } \alpha \text{ by } \beta \text{ such that } \beta \text{ is at most as complex as } \alpha\}$

2.2 The semantic treatment of the distribution puzzle

The leading idea in the existing explanation of the second key property of exceptives, namely, their limited distribution, first proposed by von Fintel (1993, 1994), is that it is semantic in nature. In other words, the meaning resulting from applying an exceptive to a quantifier with a non-universal force is ill-formed in a way. von Fintel (1993, 1994) suggested that the meaning of these constructions is necessarily contradictory. *EXH* based on innocent exclusion does not produce contradictions and has to appeal to a pragmatic constraint that the contribution of *EXH* cannot be vacuous.⁴

We will illustrate this idea by using three representative cases: an existential (an upward entailing quantifier), a downward entailing numeral, and a non-monotonic numeral. Given the assumptions outlined in the previous subsection, the LF predicted for the existential claim in (21a) is as shown in (21b).

- (21) a. *Some student but Ann passed.
 b. $[IP_2 EXH_{Alt} [IP_1 [some [student [but Ann_F]]] passed]]$

The prejacent of *EXH* gets the meaning shown in (22): some student who is not Ann passed.

- (22) $[[IP_1]]^g = T \text{ iff } \exists x[x \in \{Bill, Carl, Dan\} \ \& \ x \text{ passed}_w]$

The alternatives formed by subtracting a different individual will have the meanings shown in (23) when negated. None of these alternatives is innocently excludable, thus *EXH* cannot negate any of them. Maximally one proposition can be negated consistently with the prejacent. For example, the negation of (23a) is compatible with the prejacent: this would be a situation where Bill passed. However, in this case, the negation of (23b) or (23c) will create a contradiction. The same reasoning applies to the other propositions in (23).

- (23) a. $\lambda w. \exists x[x \in \{Ann, Carl, Dan\} \ \& \ x \text{ passed}_w]$
 b. $\lambda w. \exists x[x \in \{Ann, Bill, Dan\} \ \& \ x \text{ passed}_w]$
 c. $\lambda w. \exists x[x \in \{Ann, Bill, Carl\} \ \& \ x \text{ passed}_w]$

In order to rule this LF out, we need to impose a constraint that restricts the occurrence of *EXH* when it doesn't contribute to the sentence's meaning (Spector 2013, Gajewski 2013, Fox and Spector 2018).⁵

⁴The distribution puzzle can be resolved in the spirit of von Fintel (1993, 1994) by using a different version of *EXH*, specifically one that negates all alternatives not entailed by the prejacent (Chierchia 2013). In this case, it is predicted that the meaning of a sentence where *but* subtracts from the domain of a non-universal quantifier is necessarily contradictory. The choice between the two types of *EXH*s is arbitrary in the case of exceptives, however, as will be discussed later, the innocent exclusion based *EXH* is better suited for exceptive-additive *besides*.

⁵As observed in Hirsch's 2016 work, certain cases necessitate special consideration. For instance, when the context involves only two salient students, such as Ann and Bill, the only alternative to the prejacent in (i) will be innocently excludable, and the resulting truth conditions will be consistent.

- (i) $\lambda w. \exists x[x \in \{Bill\} \ \& \ x \text{ passed}_w]$

Hirsch (2016) suggests that this situation is disallowed by a principle that prevents the use of an existential quantifier when the conditions for using a definite article are met. This is similar to Heim's 1991 condition of Maximize Presupposition but not equivalent to it because the definite article cannot be used with *but* either, as demonstrated in (ii).

(24) NON-VACUITY: $EXH[A]$ is infelicitous if $EXH[A]$ is equivalent to A .

Because focus alternatives are obligatorily triggered and have to be used (Chierchia 2013), *but* mandates the presence of EXH making (21b) the only available LF for this sentence. However, it is blocked by the non-vacuity constraint in (24), preventing the sentence from obtaining a well-formed interpretation.

The system makes the same predictions for sentences with downward monotone numerals. To illustrate this, let's consider the predictions of the analysis for the example in (25a) with the LF in (25b).

- (25) a. *At most two students but Ann passed.
 b. $[_{IP_2} EXH_{ALT} [_{IP_1} [at\ most\ two\ [student\ [but\ Ann_F]]] passed]]$

The predicted interpretation for the prejacent is given in (26), while its alternatives are provided in their negated forms for convenience in (27). Once more, none of the alternatives will be innocently excludable. The prejacent is consistent with (27a), representing a situation where Carl, Dan, and Ann passed. However, this is incompatible with both (27b) and (27c). The same is true *mutatis mutandis* for the other negated alternatives in (26). The application of EXH is once again vacuous, and the sentence is expected to be ruled out.

(26) $[[IP_1]]^g = T$ iff $|\{Bill, Carl, Dan\} \cap \{y: y\ passed_w\}| \leq 2$

- (27) a. $\lambda w. |\{Ann, Carl, Dan\} \cap \{y: y\ passed_w\}| > 2$
 b. $\lambda w. |\{Ann, Bill, Dan\} \cap \{y: y\ passed_w\}| > 2$
 c. $\lambda w. |\{Ann, Bill, Carl\} \cap \{y: y\ passed_w\}| > 2$

In conclusion of this discussion, we will go through the final crucial case involving a non-monotonically modified numeral with *exactly* in (28).

- (28) a. *Exactly two students but/except Ann passed.
 b. $[_{IP_2} EXH_{ALT} [_{IP_1} [exactly\ two\ students\ [but\ Ann_F]] passed]]$

When examining the meaning of the prejacent in (29) and the other negated alternatives in (30), we once again notice that EXH has nothing to negate here. Let's consider a scenario where exactly two students, Carl and Dan, passed in the set $\{Bill, Carl, Dan\}$, rendering the prejacent true. This aligns with (30a) only if Ann also passed, resulting in all three individuals in $\{Ann, Carl, Dan\}$ passing. However, (30b) and (30c) cannot be true in this scenario, as there are now exactly two students who passed in both $\{Ann, Bill, Dan\}$ and $\{Ann, Bill, Carl\}$. Again, parallel considerations hold for the other two negated alternatives in (29). Since none of the alternatives is innocently excludable, the same non-vacuity principle accounts for the ungrammaticality of (28a).⁶

(29) $[[IP_1]]^{g,w} = T$ iff $|\{Bill, Carl, Dan\} \cap \{y: y\ passed_w\}| = 2$

(30) a. $\lambda w. |\{Ann, Carl, Dan\} \cap \{y: y\ passed_w\}| \neq 2$

(ii) *The student but Ann passed.

⁶Special consideration is needed to rule out (28a) in a scenario where there are three students overall: Ann, Bill, and Dan. In this particular situation, it's possible that exactly two students, either Bill and Dan, passed, while only one student each from Ann and Bill, and Ann and Dan, respectively, passed. However, this scenario can be dismissed by invoking a principle that forbids the use of *exactly two* when the entire domain of the quantifier contains precisely two individuals.

- b. $\lambda w. |\{Ann, Bill, Dan\} \cap \{y: y \text{ passed}_w\}| \neq 2$
- c. $\lambda w. |\{Ann, Bill, Carl\} \cap \{y: y \text{ passed}_w\}| \neq 2$

In conclusion, this widely used theory of exceptive constructions successfully explains two key properties of exceptives: their ability to generate the exception inference with universal and negative quantifiers and their inability to apply to quantifiers with any other quantificational force.

As we discussed in the introduction, the conditions for the use of *besides* differ from those for exceptives. In the following sections, we will argue that the same *EXH* mechanism can be adopted for the exceptive-additive construction to capture its exceptive-like uniform behavior with universal and negative quantifiers. To account for the contrast between *besides* and exceptives, particularly its compatibility with non-monotonic numerals, we propose a different way of constructing the alternatives that *EXH* utilizes in sentences containing *besides* compared to those with *but*.

2.3 *Besides* is not just domain subtraction

Given the role of *EXH* in accounting for the distribution of *but* discussed in the previous subsection and the contrast in co-occurrence restrictions between *besides* and *but*, a hypothesis regarding the freer occurrence of *besides* could be the following: while *besides* indeed shares the domain subtraction component with *but*, sentences containing *besides* – at least when it occurs with non-universal quantifiers – can lack *EXH* in their LF.⁷ This way *besides* should be able to occur with non-universal quantifiers, but there would not be any additive inference coming with such sentences.

This prediction does not align with the evidence, as demonstrated by the tests in (31) (cf. Vostrikova 2019). If the additive inferences were not obligatory when *besides* is used, the sequences in (31) would be expected to yield consistent results. This is contrary to the observed facts.

- (31) a. #Ann didn't pass. But some student besides Ann passed.
- b. #Ann didn't pass. And at most two students besides Ann passed.

Similarly, when the second sentences in (31a) and (31b) are embedded under *think*, the additive inference that Ann passed becomes an entailment of the matrix subject's doxastic state. Again, this suggests that the additive inference of *besides* with non-universal quantifiers is not optional.

- (32) a. #Edgar thinks that some student besides Ann passed, but he is not sure if Ann passed.
- b. #Edgar thinks that at most two students besides Ann passed, but he is not sure if Ann passed.

The puzzle posed by *besides* is that it interacts with universal quantifiers in a manner similar to *but*, seemingly suggesting a strengthening effect through *EXH*. However, the analysis of *but*-exceptives cannot be straightforwardly applied to *besides*, as it would incorrectly predict that *besides* should also be ungrammatical with non-universal quantifiers like *at most one* and *some*, just as *but* is.

⁷On this view *EXH* might actually be present with universal quantifiers as this would do no harm given the discussion above.

2.4 The previous treatments of *besides* and the plot

Earlier attempts to understand the semantics of exceptive-additive *besides* were successful in explaining a subset of the cases illustrated in (1)-(5). Two general approaches have been proposed: one that assumes no ambiguity in the semantics of *besides* and another that treats *besides* as a proper case of exceptive-additive ambiguity.

The first type of approach was proposed by von Fintel (1989). He provides a unified treatment of *besides* with universal quantifiers and the non-monotonic quantifier *exactly n*. The idea is that *besides* like any exceptive contributes domain subtraction. On top of this *besides* introduces the minimality condition: the subtracted set is the set with the minimal cardinality that has to be subtracted in order for the quantificational claim to be true. In other words, if a set with a smaller cardinality were subtracted, the quantificational claim would not be true.

The lexical entry proposed in (von Fintel 1989) is shown in (33): *besides* composes with a set denoted by the DP it introduces⁸, a set denoted by the restrictor, the determiner and the set denoted by the scopal argument. It subtracts the first set from the restrictor of the quantifier and adds the minimality condition (underlined in (33)).

$$(33) \quad \llbracket \textit{besides} \rrbracket^{g,w} = \lambda C_{\langle et \rangle} . \lambda A_{\langle et \rangle} . \lambda D_{\langle \langle et \rangle \langle et, t \rangle \rangle} . \lambda P_{\langle et \rangle} . D(A-C)(P) \ \& \ \underline{\forall Y [|C| > |Y| \rightarrow P \notin D(A-Y)]}$$

According to this semantics, *besides* performs two functions: it asserts a quantificational claim in which the set denoted by the first argument is subtracted from the domain (contributed by the first conjunct), and it negates all alternative quantificational claims in which a different set is subtracted, provided that these alternatives are constructed by substituting the original set with sets of smaller cardinalities (contributed by the second conjunct).

The work of the minimality condition can be observed when we analyze the specific example in (34a), which obtains the interpretation shown in (34b). The first conjunct asserts that removing Ann from the domain results in exactly two students passing. The second conjunct states that subtracting a set with a smaller cardinality than that of {Ann} will render the *exactly*-claim false. However, there is only one set with a smaller cardinality than {Ann} – the empty set. Subtracting an empty set has no domain subtraction effect, thus, we can simplify our representation of the resulting truth conditions as shown in (34c): the minimality condition essentially states that exactly two students did not pass overall. If exactly two students, other than Ann, passed, and the total number of students who passed isn't exactly two, then it follows that Ann must be the student who passed. This is because including Ann in the domain cannot reduce their number of the passing students. Therefore, combining the minimality condition with the initial quantificational claim involving domain subtraction successfully captures the additive inference the sentence in (34a) comes with.

$$(34) \quad \begin{array}{ll} \text{a.} & \text{Exactly two students besides Ann passed.} \\ \text{b.} & \llbracket (34a) \rrbracket^{g,w} = T \text{ iff } [(\{z: z \text{ is a student}_w\} - \{Ann\}) \cap \{x: x \text{ passed}_w\}] = 2 \ \& \ \underline{\forall Y [|Ann| > |Y| \rightarrow |(\{z: z \text{ is a student}_w\} - Y) \cap \{x: x \text{ passed}_w\}] \neq 2]} \\ \text{c.} & \llbracket (34a) \rrbracket^{g,w} = T \text{ iff } [(\{z: z \text{ is a student}_w\} - \{Ann\}) \cap \{x: x \text{ passed}_w\}] = 2 \ \& \ \underline{|\{z: z \text{ is a student}_w\} \cap \{x: x \text{ passed}_w\}| \neq 2}] \end{array}$$

This treatment applies to cases involving universal and negative quantifiers as well,

⁸Note that this approach assumes that the DP following an exceptive is a set denoting expression.

and it successfully explains why the inference is negative in the former and positive in the latter. For instance, when every student who is not Ann passed, but not every student passed, the logical conclusion is that Ann must be the student who did not pass. Conversely, if no student who is not Ann passed, but some student did pass, the inference is that Ann is the student who passed.

This proposal encounters a challenge when attempting to extend it to upward monotonic numerals, exemplified by *at least two* in (35) (von Fintel 1989). This approach predicts contradictory truth conditions for this sentence, since it is not possible at the same time that at least two students who are not Ann passed and that less than two students overall passed. Similarly, this approach is not applicable to existentials for the same reason.

(35) At least two students besides Ann passed.

The second type of approach is exemplified in Vostrikova (2019). This work treats the exceptive-additive flip as a proper case of ambiguity, albeit of a structural type. This analysis, however, is not straightforwardly extendable to modified numerals and such cases are not discussed.

In this paper, we advocate a no-ambiguity approach, building upon von Fintel’s unified approach to the uses of *besides* with *every* and *exactly* as our starting point. We demonstrate that this approach can be extended to all cases of modified numerals, including the upward entailing ones, by adopting several assumptions with independent theoretical motivations.

The first assumption involves the syntactic separation of the domain subtraction carried out by *besides* and the negation of alternatives performed by *EXH*, following the framework of the theory of exceptives discussed in the previous section. We argue that the only difference from exceptive *but* lies in the kind of alternatives used by *EXH*, allowing for straightforward results akin to those in von Fintel’s *exactly*-cases.

The second assumption is that the meaning of modified numerals can be decomposed into a degree quantifier and a *many*-operator within its scope, aligning with prior literature on modified numerals (Hackl 2000, Heim 2000, Mayr and Meyer 2014, Buccola and Spector 2016).

The third assumption is that *many* can be transformed into *exactly* using a maximality operator, as proposed in (Buccola and Spector 2016). Given that this step creates a syntactic constituent within the scope of a modified numeral with the meaning equivalent to *exactly*, *EXH* associated with *besides* can apply at the level of this constituent, capturing the positive inference for all modified numerals.

In what follows we spell out these assumptions and their role in explaining the interaction of *besides* with various quantifiers.

3 The proposal

3.1 *Besides* and *every*

In this section, we will present a unified treatment of *besides* with universal *every*, negative *no*, and non-monotonic *exactly-n* quantifiers using *EXH*. We will also explain the difference between exceptive-additive *besides* and pure exceptive constructions.

We build on von Fintel’s idea that similarly to exceptives the core contribution of *besides* is domain subtraction. We also adopt his proposal, which posits that both the

negative and positive inferences arise from the negation of alternative quantificational claims using different domain subtractions. Furthermore, we concur with the notion that *besides* operates with a more limited set of alternatives compared to exceptives. We implement these ideas by drawing from recent research on exceptives, which divides these functions among three distinct mechanisms: the explicit domain subtractor (in this case, *besides*), focus marking, and silent *EXH*.

More concretely, we propose that *besides* functions similarly to *but*: it performs domain subtraction and introduces a containment presupposition, as demonstrated in (36). Like *but*, *besides* combines with a plural or atomic individual (represented by the DP following *besides*) and a predicate of individuals (the restrictor set), producing a new predicate of individuals. The resulting predicate is true of atomic or plural individuals if they satisfy the restrictor predicate and do not overlap with the individual introduced by *besides*. Additionally, *besides* introduces a presupposition that the restrictor predicate holds true for the individual introduced by *besides*.

- (36) a. $\llbracket \textit{besides} \rrbracket^{g,w} = \lambda x_e. \lambda f_{\langle et \rangle}: f(x). \lambda y_e. f(y) \ \& \ \neg(x \circ y)$
 b. $\llbracket \textit{student besides Ann} \rrbracket^{g,w} = \lambda y_e. y \text{ is a student}_w \ \& \ \neg(y \circ \textit{Ann})$; is defined only if *Ann* is a student_w

The exceptive and the additive inferences are contributed by *EXH*. We propose that this operator obligatorily c-commands *besides*. Therefore, the structure of a sentence with *besides* and a universal quantifier as in (37a) looks as in (37b).

- (37) a. Every student besides Ann passed.
 b. $[_{IP_2} \text{EXH}_{ALT} [_{IP_1} \text{every} [\textit{student} [\textit{besides}_F \textit{Ann}]] \text{passed}]]$

The key difference from exceptive *but* lies in the way the alternatives are constructed. In this case, instead of placing focus on the DP following *besides* and forming alternatives by substituting this DP for a DP of at most the same complexity, we place focus on *besides* itself. We assume that the alternative for *besides* is *including*. Given the structure in (38b), the set of alternatives used by *EXH* in (37b) is as shown in (38).⁹

- (38) $\text{Alt}_{str}(\textit{every student besides}_F \textit{Ann passed}) = \left\{ \begin{array}{l} \textit{every student besides Ann passed} \\ \textit{every student including Ann passed} \end{array} \right\}$

In this case, there is only one alternative that differs from the prejacent, and it is not entailed by the prejacent, making it innocently excludable. Therefore, *EXH* negates it, while asserting the prejacent.

The overall predicted truth conditions for this sentence are as shown in (39a) If every student who is not Ann passed, but some student did not pass, then Ann is the student who did not pass. This correctly captures the negative inference (37a) comes with. Again, the inference of containment is contributed by *besides* at the presuppositional level.

- (39) a. $\llbracket (37b) \rrbracket^{g,w} = \text{T iff } \forall x [(x \text{ is a student} \ \& \ \neg(x \circ \textit{Ann})) \rightarrow x \text{ passed}_w] \ \& \ \neg \forall y [y \text{ is a student}_w \rightarrow y \text{ passed}_w]$
 defined only if *Ann* is a student_w

⁹We acknowledge that placing focus on *besides* is not always consistent with the observed prosody, with prosodic focus often showing up on the DP. One option would be to propose that the alternatives for *besides* are lexically specified. Also cf. the discussion in Section 3.4.

These assumptions alone won't deliver the correct truth conditions for a sentence where the DP following *besides* is plural, as in (40a) or (40b). The issue arises because the negation of the only alternative, essentially *every student passed*, is compatible with only one of the two individuals in the subdomain of the plural DP passing. This contradicts our intuitions about the meaning of this sentence; in fact, the exception inference applies to both individuals.

- (40) a. Every student [besides]_F Ann and Bill passed.
b. Every student [besides]_F the twins from my class passed.

In order to capture this fact, we make an additional assumption that subdomain alternatives for plurals are available independently (Bar-Lev 2021, Chatain 2021). Hence, the list of alternatives will include the alternatives where the plurality is replaced by the DPs denoting the plurality subparts as shown in (41).¹⁰

$$(41) \quad Alt_{str}(40a) = \left\{ \begin{array}{l} \text{every student besides Ann and Bill passed} \\ \text{every student besides Ann passed} \\ \text{every student besides Bill passed} \\ \text{every student including Ann and Bill passed} \end{array} \right\}$$

The overall derived truth conditions for (40a) are given in (42): the first conjunct is the prejacent that *EXH* asserts, and the second conjunct is the result of negating all other alternatives. These truth conditions capture the fact that the negative inference applies to both Ann and Bill in (40a): the prejacent, taken together with the negation of the second alternative, entails that Bill did not pass. The same reasoning applied to the third alternative guarantees that Ann did not pass. The presupposition is the inference that both Ann and Bill are students.

- (42) a. $\llbracket (40a) \rrbracket^{g,w} = \text{T}$ iff
 $\forall x[(x \text{ is a student}_w \ \& \ \neg(x \circ \text{Ann} \oplus \text{Bill})) \rightarrow x \text{ passed}_w] \ \&$
 $\exists y[(y \text{ is a student}_w \ \& \ \neg(y \circ \text{Ann})) \ \& \ \neg y \text{ passed}_w] \ \&$
 $\exists z[(z \text{ is a student}_w \ \& \ \neg(z \circ \text{Bill})) \ \& \ \neg z \text{ passed}_w] \ \&$
 $\exists x[x \text{ is a student}_w \ \& \ \neg x \text{ passed}_w]$
defined only if Ann and Bill are a students_w

To simplify the discussion, we will focus solely on cases where the individual introduced by *besides* is atomic from this point onward. However, a similar approach can always be extended to plural cases along these lines.

3.2 *Besides* and *no*

Similar to the conventional semantic approach for exceptives, this approach treats universal and negative quantifiers in a unified manner.

The alternatives for the prejacent of *EXH* in (43b) are as shown in (43c).

- (43) a. No student besides Ann passed.
b. $[IP_2 \ EXH_{ALT} \ [IP_1 \ \text{no} \ [\text{student} \ [\text{besides}_F \ \text{Ann}]] \ \text{passed} \]]$
c. $Alt_{str}(\text{no student besides}_F \ \text{Ann passed}) = \left\{ \begin{array}{l} \text{no student besides Ann passed} \\ \text{no student including Ann passed} \end{array} \right\}$

¹⁰Note that semantically alternatives of the form *Every student including Ann* or *Every student including Bill* are equivalent to the one with *including Ann and Bill*.

The predicted interpretation for the sentence is given in (44). When considering both conjuncts simultaneously, it logically follows that Ann is the student who passed. If no student other than Ann passed, but there is a student who did pass overall, that student must be Ann. This accurately captures the positive inference associated with the sentence (43a).

$$(44) \quad \begin{aligned} \llbracket (43b) \rrbracket^{g,w} = \text{T iff } & \neg \exists y [(y \text{ is a student}_w \ \& \ \neg(y \circ \text{Ann})) \ \& \ y \text{ passed}_w] \\ & \& \ \exists x [x \text{ is a student}_w \ \& \ x \text{ passed}_w] \\ & \text{defined only if Ann is a student}_w \end{aligned}$$

3.3 *Besides and exactly*

Because this approach is essentially the implementation of von Stechow's (1989) theory to *besides* in terms of *EXH*, as expected, it straightforwardly captures the positive inference contributed by *besides* when it applies to *exactly n*.

Parallel to the cases with *every* and *no* considered in the last two subsections, the sentence in (45a) gets the LF shown in (45b).

$$(45) \quad \begin{aligned} \text{a.} \quad & \text{Exactly one student besides Ann passed.} \\ \text{b.} \quad & [IP_2 \text{ EXH}_{ALT} [IP_1 [\text{exactly one student} [\text{besides}_F \text{ Ann}]] \text{ passed}]] \end{aligned}$$

Following the assumptions outlined here, the alternatives for the prejacent in (46b) are as indicated in (46). *EXH*, as in the previous cases, asserts the prejacent and negates the other alternative. To provide a preview, this leads to the positive inference that Ann passed. If exactly two students passed when we do not consider Ann, but not exactly two passed if we include her, she must be the student who passed. We will delve into the specifics of how this result is compositionally derived below.

$$(46) \quad \text{Alt}_{\text{str}}(IP_1) = \left\{ \begin{array}{l} \text{exactly two students besides Ann passed} \\ \text{exactly two students including Ann passed} \end{array} \right\}$$

We adopt the semantics for *exactly* as shown in (47a), where it is represented as a function that first composes with a degree, then with a predicate of plural individuals (its restrictor), then the resulting function composes with the second predicate of individuals (the scopal argument). This function returns *True* if and only if there exists a plurality that satisfies both predicates, has a cardinality of the degree, and no plurality with a larger cardinality satisfies the same conditions. This is implemented in (47a) by utilizing the metalinguistic *max* function defined in (47c), which applies to a predicate of degrees and returns the maximal degree satisfying that predicate if a degree satisfying the degree predicate exists, otherwise it returns 0 (cf. Hackl (2000) for a closely related treatment with *max*). The latter clause is needed to account for the *exactly zero* case, when there is no relevant plurality, thus the set of degrees is empty (Buccola and Spector 2016). In *exactly two* the *max* function applies to the predicate of degrees that selects cardinalities of the pluralities satisfying both the restrictor and the scope of *exactly two*. *Exactly two* states that the highest cardinality satisfying these conditions is two. By abstracting over the cardinality we get the meaning of *exactly* (47b).

$$(47) \quad \begin{aligned} \text{a.} \quad & \llbracket \text{exactly} \rrbracket^{g,w} = \lambda n_d. \lambda f_{\langle et \rangle}. \lambda g_{\langle et \rangle}. \max(\lambda d_d. \exists X[|X|=d \ \& \ f(X) \ \& \ g(X)]) = n \\ \text{b.} \quad & \llbracket \text{exactly two} \rrbracket^{g,w} = \lambda f_{\langle et \rangle}. \lambda g_{\langle et \rangle}. \max(\lambda d_d. \exists X[|X|=d \ \& \ f(X) \ \& \ g(X)]) = 2 \\ \text{c.} \quad & \max(P_{\langle dt \rangle}) = \iota d[P(d) \ \& \ \forall d'[P(d') \rightarrow d' \leq d]] \text{ if } \exists d[P(d)], 0 \text{ otherwise} \end{aligned}$$

The prejacent of *EXH* gets the following truth conditions: it is true if and only if there are precisely two students, other than Ann, who passed.

$$(48) \quad \llbracket IP_1 \rrbracket^{g,w} = T \text{ iff } \max(\lambda d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = 2$$

EXH asserts its prejacent and negates the only alternative in (46) that is different from it. This results in the following overall truth conditions: the sentence is predicted to be true if and only if there was precisely one student, excluding Ann, who passed, but not exactly one student passed in total. Since including Ann in the group of students couldn't have reduced the total number of passing students compared to when Ann was excluded, this can only hold if Ann indeed passed.

$$(49) \quad \llbracket (45b) \rrbracket^{g,w} = T \text{ iff } \max(\lambda d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = 2 \\ \& \ \max(\lambda m. \exists Y[|Y|=m \ \& \ Y \text{ is a student}_w \ \& \ Y \text{ passed}_w]) \neq 2 \\ \text{defined only if Ann is a student}_w$$

This discussion makes the difference between exceptive constructions and *besides* clear. As demonstrated in Section 2, exceptive constructions such as *but* and *except* are predicted to be incompatible with *exactly* due to a distinct method of constructing alternatives. Specifically, in exceptive constructions, the only element marked with focus is the DP following the exceptive marker. Hence, the alternatives vary only with respect to the elements subtracted from the domain of the quantifier. The alternative where nothing is subtracted is not included in the set of alternatives. The presence of such an alternative is crucial for the proposed account of the interaction between *besides* and *exactly*.

As the reader can observe, we propose to derive both the positive inference of *besides* with negative quantifiers and *exactly n* numerals and the negative inference with *every* using the same mechanism. We do not posit any ambiguity, whether it's structural or lexical, in the interpretation of *besides*. Instead, the nature of the inference it conveys is determined by the properties of the quantifier whose domain *besides* restricts.

The challenge this approach faces is that the additive inference is observed not only with downward entailing *no* and non-monotonic *exactly n*, but also with upward entailing quantifiers as well, such as existential *some* and *at least n* numerals. In the following sections, we will delve into this challenge and present our proposed solution.

Before exploring this extension of our proposal, we will briefly consider an alternative approach to focus marking and constructing the alternatives used by *EXH* associated with *besides*, which might result in the same list of alternatives.

3.4 Structurally simpler alternatives only?

Another approach to constructing alternatives that might lead to the same overall meaning for the sentence is worth considering here. In this alternative approach, one could argue that the entire *besides*-phrase is focused, as shown in (50a), and the list of alternatives includes structurally simpler alternatives. For *EXH* in (50a), there is only one structurally simpler alternative, namely the one in (50b). Its meaning is equivalent to the one of *Exactly one student including Ann passed*. As we explored in the previous section, negating this alternative while asserting the prejacent results in the additive inference and correctly captures the meaning of the sentence. This result would thus

extend to (50a) with F-marking of the entire *besides*-phrase.¹¹

- (50) a. EXH_{Alt} Exactly one student [besides Ann]_F passed
b. Exactly one student passed.

We will demonstrate the necessity of restricting the list of alternatives to structurally simpler ones, avoiding equally complex alternatives. These are alternatives that differ in terms of the DP following *besides*, such as *Exactly one student besides Bill passed*. This means that relying solely on the structural theory of focus alternatives isn't sufficient to obtain the correct set of alternatives. Furthermore, we will illustrate that even with this restriction, an additional assumption is still necessary: that subdomain alternatives for plurals are independently available.

In the simple case, like the one considered in (50a), alternatives of equal complexity could be included in the set of alternatives EXH operates on, along with strictly simpler alternatives. Let's assume that the relevant students are Ann, Bill, and Carl. Then the meanings of the alternatives are as shown in (51). Here, only the alternative provided in (51d) is present in every maximal set of excludable alternatives, as shown in (52). Thus, it's the only alternative that can be innocently excluded and negated by EXH.

- (51) a. $\lambda w. \llbracket \text{exactly 1 student besides Ann passed} \rrbracket^w = \lambda w. |\{b, c\} \cap \{x: x \text{ passed}_w\}| = 1$
b. $\lambda w. \llbracket \text{exactly 1 student besides Bill passed} \rrbracket^w = \lambda w. |\{a, c\} \cap \{x: x \text{ passed}_w\}| = 1$
c. $\lambda w. \llbracket \text{exactly 1 student besides Carl passed} \rrbracket^w = \lambda w. |\{a, b\} \cap \{x: x \text{ passed}_w\}| = 1$
d. $\lambda w. \llbracket \text{exactly 1 student passed} \rrbracket^w = \lambda w. |\{a, b, c\} \cap \{x: x \text{ passed}_w\}| = 1$
(52) a. Max₁ (The situation where Bill and Ann passed): $\{(51b), (51d)\}$
b. Max₂ (The situation where Carl and Ann passed): $\{(51c), (51d)\}$

The reason why equally complex alternatives should not be generally included in the set of alternatives EXH operates on has to do with *besides*-phrases hosting plural DPs, like in (53). Here, including such alternatives would incorrectly predict that the positive inference applies to only one of the individuals introduced by the plural following *besides*, leading to the inference that Ann or Bill passed in the case of (53).

- (53) Exactly one student [besides Ann and Bill]_F passed.

Let's illustrate this with a scenario involving four individuals: Ann, Bill, Carl, and Dan. In this context, the intensionalized prejacent's meaning is as shown in (54a), and the meanings of the other alternatives are given in (54b)-(54k). The maximal sets of alternatives that can be negated together with the assertion of the prejacent are given in (55). The only alternative consistently present in all sets is the one in (54k). This is the only innocently excludable alternative, meaning that EXH can only negate this specific alternative.

- (54) a. $\lambda w. \llbracket \text{exactly 1 student besides A\&B pass} \rrbracket^w = \lambda w. |\{c, d\} \cap \{x: x \text{ passed}_w\}| = 1$
b. $\lambda w. \llbracket \text{exactly 1 student besides B\&C pass} \rrbracket^w = \lambda w. |\{a, d\} \cap \{x: x \text{ passed}_w\}| = 1$
c. $\lambda w. \llbracket \text{exactly 1 student besides B\&D pass} \rrbracket^w = \lambda w. |\{a, c\} \cap \{x: x \text{ passed}_w\}| = 1$
d. $\lambda w. \llbracket \text{exactly 1 student besides A\&C pass} \rrbracket^w = \lambda w. |\{b, d\} \cap \{x: x \text{ passed}_w\}| = 1$

¹¹One could rely on F-projection (Selkirk 1996, Schwarzschild 1999 a.m.o.) to get this. The idea would be that it is the DP that is F-marked – attracting pitch accent according to something like the nuclear stress rule – and *besides* gets F-marked because the former is the internal argument of the latter. This in turn allows the whole *besides*-phrase to be F-marked. Since the *besides*-phrase is not the argument of the restrictor phrase no further F-projection takes place.

- e. $\lambda w. \llbracket \text{exactly 1 student besides A\&D pass} \rrbracket^w = \lambda w. |\{b,c\} \cap \{x: x \text{ passed}_w\}| = 1$
 - f. $\lambda w. \llbracket \text{exactly 1 student besides C\&D pass} \rrbracket^w = \lambda w. |\{a,b\} \cap \{x: x \text{ passed}_w\}| = 1$
 - g. $\lambda w. \llbracket \text{exactly 1 student besides Bill pass} \rrbracket^w = \lambda w. |\{a,c,d\} \cap \{x: x \text{ passed}_w\}| = 1$
 - h. $\lambda w. \llbracket \text{exactly 1 student besides Ann pass} \rrbracket^w = \lambda w. |\{b,c,d\} \cap \{x: x \text{ passed}_w\}| = 1$
 - i. $\lambda w. \llbracket \text{exactly 1 student besides Carl pass} \rrbracket^w = \lambda w. |\{a,b,d\} \cap \{x: x \text{ passed}_w\}| = 1$
 - j. $\lambda w. \llbracket \text{exactly 1 student besides Dan pass} \rrbracket^w = \lambda w. |\{a,b,c\} \cap \{x: x \text{ passed}_w\}| = 1$
 - k. $\lambda w. \llbracket \text{exactly 1 student passed} \rrbracket^w = \lambda w. |\{a,b,c,d\} \cap \{x: x \text{ passed}_w\}| = 1$
- (55)
- a. Max_1 (Carl and Ann passed): $\{(54c), (54d), (54g), (54j), (54k)\}$
 - b. Max_2 (Dan and Ann passed): $\{(54b), (54e), (54g), (54i), (54k)\}$
 - c. Max_3 (Carl and Bill passed): $\{(54b), (54e), (54h), (54j), (54k)\}$
 - d. Max_4 (Dan and Bill passed): $\{(54d), (54c), (54h), (54i), (54k)\}$
 - e. Max_4 (Dan, Bill, and Ann passed): $\{(54b), (54d), (54g), (54h), (54i), (54j), (54f), (54k)\}$
 - f. Max_5 (Carl, Bill, and Ann passed): $\{(54c), (54e), (54g), (54h), (54i), (54j), (54f), (54k)\}$

The predicted resulting interpretation is given in (56). If exactly one among Carl and Dan passed, but not exactly one student passed overall, this means that either Ann or Bill also passed, but not necessarily both of them.

$$(56) \quad \llbracket (53) \rrbracket^{g,w} = T \text{ iff } |\{c,d\} \cap \{x: x \text{ passed}_w\}| = 1 \ \& \ |\{a,b,c,d\} \cap \{x: x \text{ passed}_w\}| \neq 1$$

It is quite clear what we need to do to get the additive inference right: we want to negate the alternatives in (54g) (*Exactly one student besides Bill passed*) and (54h) (*Exactly one student besides Ann passed*). Both of those alternatives are structurally simpler than the prejacent of *EXH* in (53).

Given these observations, one might be tempted to conclude that it is sufficient to place a restriction on the set of alternatives used by *EXH*. Specifically, this restriction would state that only structurally simpler alternatives are included—those derived by removing parts of the syntactic tree. Alternatives generated by substituting lexical items with other lexical items would, therefore, be excluded.

However, such an approach would fall short of deriving the correct meaning for the sentence in (57). The issue is that while with pluralities formed by conjunction the subdomain alternatives are structurally simpler, this is not the case for other plurals. It is not clear how to derive the expressions referring to individual twins by simplifying *the twins from my class*.

$$(57) \quad \text{Exactly one student } [\text{besides the twins from my class}]_F \text{ passed}$$

Therefore, even if we deal with a restricted set of structurally simpler alternatives, we still require subdomain alternatives for plurals to be independently accessible if we adopt this method of F-marking and alternative construction. In the following we will revert to representations with F-marking *besides* only.

4 *Besides* and upward and downward entailing modified numerals

This section introduces the tools necessary to extend our account to other cases of modified numerals, focusing on upward entailing numerals.

4.1 The challenge of extending the approach with *EXH* to upward entailing quantifiers

The proposed method combines a quantificational claim with domain subtraction and the negation of this claim over a larger domain. The reason this approach does not straightforwardly extend to contexts where *besides* is in the restrictor of an upward-entailing quantifier is evident: if the only alternative that differs from the prejacent is also entailed by it, it cannot be negated without conflicting with the assertion of the prejacent itself.¹² We will illustrate this point with the example in (58).

(58) At least two students [*besides_F* Ann] passed.

Given the assumptions outlined thus far, (59a) represents a possible LF for the sentence in question. Consequently, the alternatives engaged by *EXH* would be as shown in (59b). Considering that both alternatives in (59b) are entailed by the prejacent, neither is innocently excludable. Therefore, *EXH* has no alternatives to negate in this case.

- (59) a. $[IP_2 \text{ EXH}_{ALT} [IP_1 [\text{at least two students } [\text{besides}_F \text{ Ann}]] \text{ passed }]]$
 b. $Alt_{str}(IP_1) = \left\{ \begin{array}{l} \text{at least two students besides Ann passed} \\ \text{at least two students including Ann passed} \end{array} \right\}$

Given that *EXH* does not contribute additional meaning to the sentence, this LF is ruled out by the same non-vacuity constraint that we invoked to explain the distribution of exceptive constructions. However, given that the sentence in (58) is perfectly grammatical and that *besides* contributes an additive meaning, there must exist an alternative LF that accounts for this interpretation.

4.2 The decompositional approach to modified numerals

Our solution to the puzzle posed by the interaction of upward entailing quantifiers and *besides* is based on the theory that modified numerals are quantifiers over degrees, requiring quantifier raising for their interpretation (see Hackl 2000, Mayr and Meyer 2014, Buccola and Spector 2016). This theory extends the idea of treating degree constructions quantificationally, as originally suggested in (Heim 2000), to modified numerals.

We propose that when such a quantifier moves, it can optionally leave behind a constituent with the meaning *exactly*. This means that we can directly apply our approach for *exactly n* numerals to these cases by placing *EXH* below the scope of the upward-entailing numeral itself. We will first outline our assumptions regarding the interpretation of sentences containing modified numerals such as *at least two*, and then

¹²The interaction of upward entailing quantifiers with *besides* is the reason why the innocent exclusion variant of *EXH* is more fitting for this construction. The contradiction-inducing *EXH* negates those alternatives that the prejacent does not entail. In the case at hand, *EXH* finds itself with nothing to negate, given that the sole alternative that differs from the prejacent is, in fact, entailed by it. Consequently, we must resort to an alternative mechanism to dismiss (59a). One option is to adopt the non-vacuity constraint, as we did in this paper. Nevertheless, imposing this restriction into the contradiction-generating *EXH* presents a tension with other instances where this form of *EXH* is customarily applied. For instance, the contradiction-based account of NPI distribution is incompatible with the non-vacuity constraint. According to this analysis, the NPI *any* triggers subdomain alternatives that must be utilized by *EXH*. If *any* is not in a downward-entailing (DE) environment, the application of *EXH* leads to a contradiction. Crucially, however, if *any* is in a DE environment, the application of *EXH* is vacuous, resulting in a well-formed meaning (Chierchia 2013).

demonstrate how these assumptions naturally account for their interaction with *besides*. Our starting point is the treatment of modified numerals as comparative degree quantifiers proposed by (Hackl 2000). This treatment has three main components: a silent degree function *many*, a degree denoted by the numeral, and a degree quantifier denoted by the numeral itself.¹³

We assume that *at least two* is a quantifier over degrees with the semantics given in (60a): like any degree quantifier, it seeks to compose with a predicate of degrees, an expression of type $\langle dt \rangle$. It introduces existential quantification over degrees and returns Truth if there is a degree that satisfies this predicate that is equal to two or larger. Accordingly, abstracting over the degree two, we get the semantics of *at least* in (60b). Thus, *at least two* does not compose with the plural predicate *students* directly.

- (60) a. $\llbracket \text{at least two} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \exists d[d \geq 2 \ \& \ f(d)]$
 b. $\llbracket \text{at least} \rrbracket^{g,w} = \lambda n_d. \lambda f_{\langle dt \rangle}. \exists d[d \geq n \ \& \ f(d)]$

We adhere to the conventional degree semantics assumption that a degree quantifier is a sister of a silent operator *many* of type $\langle d, \langle \langle et \rangle, \langle et \rangle, t \rangle \rangle$. Due to the type mismatch, *at least two* cannot be interpreted in its initial position. It undergoes quantifier raising, leaving behind a trace of type d . A numerical index, matching the index on the trace, is then merged below the landing site of *at least two*.

The meaning of *many* is given in (61c). It is a function that first composes with a degree and subsequently with two predicates of individuals and returns Truth if and only if there is a plurality that satisfies both predicates with the cardinality of the degree. The trace left by *at least two* serves as the first argument of the silent *many*.

- (61) a. At least two students passed.
 b. $[_{IP_2} \text{ at least two } [_{IP_1} 1 \ [\text{many } d_1 \text{ students passed}]]]$
 c. $\llbracket \text{many} \rrbracket^{g,w} = \lambda n_d. \lambda f_{\langle et \rangle}. \lambda g_{\langle et \rangle}. \exists X[|X|=n \ \& \ f(X) \ \& \ g(X)]$

These assumptions are sufficient to interpret this sentence that does not have *besides*. The sister of *at least two* is the predicate of degrees shown in (62a). Accordingly, the overall predicted truth conditions for this sentence are given in (62b).

- (62) a. $\llbracket IP_1 \rrbracket^{g,w} = \lambda n_d. \exists X[|X|=n \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]$
 b. $\llbracket (61b) \rrbracket^{g,w} = T \text{ iff } \exists d[d \geq 2 \ \& \ \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]]$

4.3 Turning *many* into *exactly*

By adopting the compositional analysis we created an additional potential position for *EXH* associated with *besides* below the scope of the modified numeral. These assumptions alone, however, are not sufficient to account for the interaction of *at least two* and *besides*. There is no downward entailing or non-monotonic quantifier in the LF in (61b).

An attempt to insert *EXH* above *many* would still lead to a vacuous application of *EXH*. This is because *many*, being an existential quantifier, is also an upward-entailing quantifier.

- (63) $*[_{IP_4} \text{ at least two } [_{IP_3} 1 \ [_{IP_2} \text{ EXH}_{ALT} \ [_{IP_1} \text{ many } d_1 \text{ students besides Ann passed}]]]]]$

¹³Hackl (2000) extends this approach to *exactly n*-quantifiers. In principle this would also be compatible with the present proposal, but we do not discuss this here.

We suggest that an additional element is at work in this context: a silent operator that transforms *many* into *exactly*. We propose that this additional element is inserted in the LF to prevent a contradiction and create a well-formed meaning for the sentence. We will adopt the proposal by Buccola and Spector (2016), where the constituent with the meaning *exactly* is created via a type-shifter containing a *max* function applied to a trace left behind by the modified numeral.

The proposal in (Buccola and Spector 2016) posits that modified numerals can undergo quantifier raising twice, as illustrated in the LF below. The first movement is already familiar from the previous discussion. The trace d_2 left behind by the second movement type-shifted through the shifter *MAX* presented in (65), which employs the same metalinguistic *max* function used above in the definition of *exactly*. The *EXH* associated with *besides* is merged above the first landing site of the modified numeral.

$$(64) \quad [IP_6 \text{ at least one } [IP_5 2 [IP_4 EXH_{ALT} [IP_3 d_{2MAX} [IP_2 1 [IP_1 \text{ many } d_1 \text{ student besides}_F \text{ Ann passed}]]]]]]]$$

$$(65) \quad \llbracket MAX \rrbracket^{g,w} = \lambda d_d. \lambda P_{\langle dt \rangle}. \max(P)=d$$

With these assumptions, there exists a constituent, specifically IP_3 , whose meaning is equivalent with that of the *exactly*-claim: excluding Ann, the number of students who passed is exactly the degree that the assignment function g yields when it is applied to the index 2¹⁴.

$$(66) \quad \llbracket IP_3 \rrbracket^{g,w} = T \text{ iff } \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = g(2)$$

Building on our previous discussion regarding the interaction between *besides* and *exactly*, we can straightforwardly apply *EXH* to this constituent to obtain the additive meaning. To spell it out, the sole alternative distinct from the prejacent is given in (67a). The intensionalized meaning of this alternative is provided below in (67b).

$$(67) \quad \begin{array}{ll} \text{a.} & [IP_3 d_{2MAX} [IP_2 1 [IP_1 \text{ many } d_1 \text{ student including Ann passed}]]]]]] \\ \text{b.} & \lambda w. \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]) = g(2) \end{array}$$

Consequently, the modified numeral *at least two* composes with the predicate in (68): this predicate is true of a degree if this degree is the number of students who passed if Ann is not taken into account, but it is not the overall number of students who passed.

$$(68) \quad \llbracket IP_5 \rrbracket^{g,w} = \lambda n. \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = n \\ \& \ \neg \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]) = n$$

¹⁴Alternatively, the same outcome can potentially be achieved without the need for a distinct type-shifting mechanism or a dedicated maximality operator embedded within the structure. Instead, an additional instance of *EXH* can be inserted beneath the scope of the *EXH* associated with *besides*, as illustrated in (i).

$$(i) \quad [IP_5 \text{ at least two } [IP_4 1 [IP_3 EXH_{ALT_2} [IP_2 EXH_{ALT_1} [IP_1 \text{ many } d_{1F_1} \text{ student besides}_{F_2} \text{ Ann passed}]]]]]]]$$

The alternatives for this additional operator are constructed by substituting the trace with its focus alternatives. This process is akin to generating the upper bound meaning for bare numerals, such as *two students*. One important thing to have in mind though is that the result of this will be equivalent to the result achieved with the maximality operator only under the assumption that there is a null individual. Otherwise, the prediction is that all numerals with existential semantics would come with an existence entailment. This will run into problems with DE numerals that will be discussed later in this chapter.

Finally, the overall meaning of the sentence is as shown in (69). The sentence is predicted to be true if and only if there exists a degree equal to 2 or larger such that this is the number of students who passed when Ann is not taken into consideration, yet it is not the total number of students who passed. This accurately captures the additive inference: these truth conditions can only be met if Ann passed. To illustrate, let us consider the situation where the opposite occurred: Ann did not pass. In this case, the number of students who passed remains unchanged, regardless of whether we include Ann. Thus, the truth conditions will not be satisfied in this scenario.

$$(69) \quad \llbracket (64) \rrbracket^{g,w} = T \text{ iff } \exists n[n \geq 2 \ \& \ \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = n \ \& \ \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]) \neq n]$$

Our approach is based on the intuition that there is always an exact number of people who passed. When we choose to describe this number with the modified numeral *at least two*, we are not providing this number directly – presumably for reasons of ignorance – but describing it indirectly via comparison with another number, namely two.

It is standard to incorporate maximality directly into the meaning of a degree operator, in our case that would be the modified numeral itself (Heim 2000). Our proposal is to detach the maximality operator from the degree quantifier.

It’s noteworthy, however, that achieving the correct interpretation for upward entailing modified numerals doesn’t necessitate maximality at all. The concept of type-shifting, which includes a built-in maximality function, was initially introduced by Buccola and Spector (2016) primarily for downward entailing numerals. Considering that this type-shifting mechanism was suggested for a different category of modified numerals, we posit its availability for various types of numerals as required. To enable the grammaticality of *besides* in this context, the upward entailing constituent within the scope of *at least two* must undergo the transformations we’ve described to become non-monotonic.

Another type of upward-entailing numeral – *more than n* in (70a) – is handled in an entirely parallel manner. The sole distinction from upward numerals like *at least n* lies in the degree quantifier’s introduced relation: it is more than two. Consequently, the anticipated truth conditions for (70a) are depicted in (70b). As readers can verify, these truth conditions necessitate Ann’s is passing the test, effectively capturing the additive inference.

$$(70) \quad \begin{array}{ll} \text{a.} & \text{More than two students besides Ann passed.} \\ \text{b.} & \llbracket (70a) \rrbracket^{g,w} = T \text{ iff } \exists n[n > 2 \ \& \ \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w]) = n \ \& \ \neg \max(\lambda d_d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]) = n] \end{array}$$

Upward-entailing modified numerals are not the sole upward-entailing quantifiers that are compatible with *besides*. It can also occur with existentials and with *several*, as demonstrated below. Crucially, additive inferences are licensed here as well.

$$(71) \quad \begin{array}{ll} \text{a.} & \text{Some student(s) besides Ann passed.} \\ \text{b.} & \text{Several students besides Ann passed.} \end{array}$$

Placing *EXH* above the existential will not produce the desired result for the reasons discussed above. To account for the additive readings of *besides* in these contexts, we suggest that existentials can optionally be interpreted as degree quantifiers. In the

singular case, this interpretation is equivalent to *at least one* as demonstrated in (72a), while in the plural case, it's equivalent to *more than one*, as illustrated in (72b). The latter is also the interpretation of *several*.

- (72) a. $\llbracket \text{some} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \exists d[d \geq 1 \ \& \ f(d)]$
 b. $\llbracket \text{some}' \rrbracket^{g,w} = \llbracket \text{several} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \exists d[d > 1 \ \& \ f(d)]$

Under this proposal, *some* undergoes QR similar to any degree quantifier, leaving behind *many* and the trace. The same mechanism can be used to transform the upward-entailing constituent into a non-monotonic one beneath the scope of *EXH*.

By adopting the decomposition approach to modified numerals and existentials, where they are considered quantifiers over degrees undergoing QR, thus creating a scopal opportunity for *EXH* beneath the scope of the upward entailing quantifier itself, we have accurately accounted for the grammaticality of *besides* in sentences featuring such quantifiers and have successfully explained its additive reading in these cases.

4.4 *Besides* and downward monotonic numerals

In this section we will explore the interaction of downward entailing modified numerals and *besides*.

4.5 Extending the proposed approach to DE numerals

The decompositional account outlined in the previous section can be readily extended to downward modified numerals like *fewer than n* and *at most n*. In line with Buccola and Spector (2016), we adopt an approach that treats them as existential quantifiers over degrees, as illustrated in (73) and (74). This treatment parallels how we handle *at least n* and *more than n*.

- (73) a. $\llbracket \text{fewer than} \rrbracket^{g,w} = \lambda n_d. \lambda f_{\langle dt \rangle}. \exists m[m < n \ \& \ f(m)]$
 b. $\llbracket \text{fewer than two} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \exists m[m < 2 \ \& \ f(m)]$
 (74) a. $\llbracket \text{at most} \rrbracket^{g,w} = \lambda n_d. \lambda f_{\langle dt \rangle}. \exists m[m \leq n \ \& \ f(m)]$
 b. $\llbracket \text{at most two} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \exists m[m \leq 2 \ \& \ f(m)]$

This approach differs from the more conventional treatment of downward-entailing numerals, where they are considered negative degree quantifiers (e.g. Hackl 2000), as exemplified in (75). However, if the maximality operator is present in the degree predicate that the modified numeral combines with, the existential semantics in (74) yield the same truth conditions as those obtained by applying the lexical entries in (75) (where they are treated as negative quantifiers) to a constituent lacking the maximality operator (Buccola and Spector 2016).

- (75) a. $\llbracket \text{fewer than two} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \neg \exists m[m \geq 2 \ \& \ f(m)]$
 b. $\llbracket \text{at most two} \rrbracket^{g,w} = \lambda f_{\langle dt \rangle}. \neg \exists m[m > 2 \ \& \ f(m)]$

To illustrate this point using a specific example, let's examine (76) and the two formulations of its truth conditions in (77). The assumption that $\max(\emptyset)=0$, which we have adopted following (Buccola and Spector 2016), plays a crucial role here: it allows avoiding the existential entailment that some students passed in (77b), making the two formulations equivalent. Both accurately predict the sentence to be true when 0, 1, or 2 students have passed and false when 3 or more have passed. The delimitative aspect

typically associated with the modified numeral itself is integrated into the maximality function. When three or more students have passed, the maximal cardinality for which a plurality of students who passed exists is not smaller or equal to 2.

(76) At most two students passed.

- (77) a. $\neg\exists d[d > 2 \ \& \ \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w]]$
 b. $\exists d[d \leq 2 \ \& \ \max(\lambda n. \exists X[|X|=n \ \& \ X \text{ is a student}_w \ \& \ X \text{ passed}_w])=d]$

It's important to highlight that here, unlike in the previously discussed cases with upward-entailing numerals, the presence of the maximality operator is essential to obtain the correct interpretation for sentences containing these downward-entailing numerals under the assumption that they have the existential semantics shown in (73a) and (74a). That is, the maximality operator is required even in the absence of *besides*.

Taking into account the discussion in the previous section, we propose that (78a) gets the LF shown in (78b). (Here, we are using the type-shifter with built-in maximality as suggested by Buccola and Spector (2016), but a similar outcome could be achieved with an additional *EXH*, as shown above.)

- (78) a. At most two students besides Ann passed.
 b. $[IP_6 \text{ at most two } [IP_5 \ 2 \ [IP_4 \ EXH_{ALT} [IP_3 \ d_{2MAX} [IP_2 \ 1 \ [IP_1 \ \text{many } d_1 \ \text{student besides}_F \text{ Ann passed}]]]]]]]$

The sister of *at most two* has the denotation as presented in (79). This is the predicate of degrees that holds true of a degree if precisely this number of students passed without considering Ann, and not this number of students passed in total.

- (79) $\llbracket IP_2 \rrbracket^{g,w} = \lambda n. \max(\lambda d. \exists x[|x|=d \ \& \ x \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w])=n \ \& \ \max(\lambda m. \exists Y[|Y|=m \ \& \ Y \text{ is a student}_w \ \& \ Y \text{ passed}_w]) \neq n$

The overall truth conditions for (78a) are given in (80). The sentence is predicted to be true if and only if there exists a number, which represents the maximal count of students who passed without considering Ann, and this number is distinct from the total number of students who passed, and it is less than or equal to two. This accurately captures the additive inference that Ann passed associated with the sentence. Let's contemplate the scenario in which Ann did not pass, and the total count of passing students is 1. Based on these truth conditions, the sentence is predicted to be false in this context. This is because the number of students who passed remains unchanged, whether we include Ann or not. Consequently, there is no degree that satisfies both the restrictor and the scope of the existential quantifier.

- (80) $\llbracket (78a) \rrbracket^{w,g} = T \text{ iff } \exists n[n \leq 2 \ \& \ \max(\lambda d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w])=n \ \& \ \max(\lambda m. \exists Y[|Y|=m \ \& \ Y \text{ is a student}_w \ \& \ Y \text{ passed}_w]) \neq n]$

Downward entailing numerals of the form *fewer than n* are treated in a parallel manner. To illustrate, the predicted truth conditions for (81a) are given in (81b). As the reader can verify, these truth conditions necessitate that Ann passed, accurately capturing the additive inference. The only difference to the previous case lies in the relation introduced by the numeral: the cardinality existentially quantified over is less than two.

- (81) a. Fewer than two students besides Ann passed.

- b. $\llbracket (81a) \rrbracket^{w,g} = \text{T}$ iff $\exists n[n < 2 \ \& \ \max(\lambda d. \exists X[|X|=d \ \& \ X \text{ is a student}_w \ \& \ \neg(X \circ \text{Ann}) \ \& \ X \text{ passed}_w])=n \ \& \ \max(\lambda m. \exists Y[|Y|=m \ \& \ Y \text{ is a student}_w \ \& \ Y \text{ passed}_w]) \neq n$

5 The motivation for separation of the domain subtraction and the minimality condition

We have built upon the existing theories of exceptive constructions, which distinguish between domain subtraction and the negation of alternatives at a syntactic level (Gajewski 2008, 2013, Hirsch 2016, Crnić 2021), and we proposed to extend this approach to exceptive-additive *besides*.

As is evident from our discussion of modified numerals, the separation of these two steps is crucial for our theory to work out. This is because the quantifier to which *besides* provides the restriction in such cases is upward entailing *many*. If *besides* were to incorporate the minimality condition, it would lead to ungrammaticality at the level of that constituent. By allowing the minimality condition to be contributed by a separate element, we can apply it at the stage when this upward entailing constituent is transformed into a non-monotonic one. Now, we will discuss whether this choice has independent motivation.

Three arguments have been put forward in support of separating domain subtraction from the negation of alternatives in the analysis of exceptives like *but*. Two of these arguments involve NPIs (Gajewski 2008), and the third pertains to cases where modal operators intervene between the modal and *EXH* (Crnić 2021). We will discuss these arguments and demonstrate that two of them can be extended to *besides*.

The argument that cannot be straightforwardly extended to *besides* concerns the interaction between *but* and the NPI *any*. It is standard to assume that *any* has an existential semantics and is licensed in downward entailing environments. In (82a), *but* provides the exception for the negative claim resulting from the combination of the existential NPI *any* and negation. If *but* itself contributed negation of alternatives in addition to domain subtraction, the sentence should come out ungrammatical, just like the one in (82b). The reason is that *but* would restrict the scope of a non-universal quantifier, which we have seen to lead to ungrammaticality. The LF where *EXH* is placed above negation, as shown in (82c), however delivers the correct result. The reason for this is that the existential *any* under negation obtains an interpretation equivalent to the negative universal quantifier.

- (82) a. I did not eat any vegetables but cucumbers.
 b. *I ate some vegetables but cucumbers.
 c. $[\text{EXH}_{Alt} [\text{NEG I ate any vegetables but cucumbers}_F]]$

While it is true that *besides*, like *but*, can occur in this position and, according to the analysis we proposed, *EXH* must be merged above negation, a theory-independent argument cannot be developed in this case. Unlike *but*-exceptives, *besides* can grammatically occur with existential quantifiers (as illustrated in (83c)).

- (83) a. I did not eat any vegetables besides cucumbers.
 b. $[\text{EXH}_{Alt} [\text{NEG I ate any vegetables} [\text{besides cucumbers}_F]]]$
 c. I ate some vegetables besides cucumbers.

The second argument relates to the licensing of NPIs within the scope of negative

quantifiers with *but*-phrases in their restrictor. This argument can be extended to *besides*-phrases. As demonstrated in (84), NPIs are licensed in this environment. If the *but*-phrase were to contribute, in addition to domain subtraction, the negation of alternative quantificational claims that differ from the original only in terms of what is subtracted, the scope of the negative quantifier in (84) would not constitute a downward-entailing (DE) environment. This is because negating a negative claim results in a positive existential claim, placing the NPI in an upward-entailing (UE) environment.

(84) No one but John ate any cucumbers.

Empirically, this can be appreciated by looking at the entailment pattern for the whole sentence and observing that the position of *any* is not in a DE environment globally. (86a) does not entail (86b). The former can be true in a scenario where no one other than John ate any vegetables, and John ate only tomatoes. The latter is false in that scenario.

- (85) a. No one but John ate vegetables.
b. No one but John ate cucumbers.

One clarification is due here. Not only weak NPIs like *any* are licensed in this environment, but also strong NPIs like *in years*, as shown in (86a) (Gajewski 2008). This is important because weak NPIs are known to be sensitive to the truth-conditional content only and can ignore presuppositions that disrupt the monotonicity pattern (von Stechow 1999). Focusing on the weak NPI pattern in (84) leaves the possibility open that the negation of the alternatives is introduced at the presuppositional level and this is why it does not intervene with the licensing. This, however, is not an option for the strong NPI in (86a), as they are sensitive to the presuppositional content as well (Gajewski 2011). This is shown by the contrast between (86b) and (86c): *only* presupposes its prejacent, which is a positive statement. This does not disrupt/affect the NPI licensing of *any* in (86c), but blocks *in years* in (86b).

- (86) a. No one but John has visited me in years.
b. *Only John has visited me in years.
c. Only John ate any vegetables.

Strong NPI licensing in the scope of *no one* in (86a) can be accounted for by separating domain subtraction and exclusion of alternatives syntactically, as shown in (87). Although the NPI is not in a DE environment globally, there exists a syntactic constituent, namely the sister of *EXH* where the requirements for NPI licensing are satisfied. This is sufficient for NPI licensing.

(87) [*EXH*_{Alt} [no one but John has visited me in years]]

In exactly the same way, NPIs – weak and strong – are licensed in the scope of *no one* whose domain is modified by *besides*, as shown in (88). However, if negation of the alternatives were built into the meaning of *besides* this would be unexpected, as the environment of *any* and *in years* would not be predicted to be monotonic.

- (88) a. No one besides John ate any cucumbers.
b. No one besides John has visited me in years.

Empirically this can be appreciated by observing that (89a) does not entail (89b). (89a)

is true in the scenario where no one other than John visited me last year and John visited me during the winter only. (89b) is false in this scenario.

- (89) a. No one besides John visited me last year.
b. No one besides John visited me last spring.

The syntactic separation of domain subtraction and minimality, as shown in (90), resolves the problem and provides an explanation for NPI licensing. The NPI is licensed locally inside the sister of *EXH*. Within this constituent, it is in the scope of the negative quantifier and is thus in a DE environment.

- (90) [*EXH*_{ALT} [no one [besides John]_F has visited me in years]]

We illustrate the third argument through the modal suspension readings of exceptives. Crnić (2021) reports a contrast between (91b) and (91c): while a sentence containing an exceptive cannot be conjoined with another sentence that directly contradicts the exceptive inference, native speakers of English find it acceptable to conjoin it with a modalized version of that sentence.

- (91) a. Every student except Dina arrived, ...
b. #and even she did!
c. and perhaps even she did!

Crnić (2021) proposes that this fact can be derived under the assumption that a silent modal operator contributing to the speaker's certainty can intervene between *EXH* and the universal statement. Accordingly, the resulting reading is: 'The speaker is certain that every student who is not Dina arrived, and he allows the possibility that not every student who is not Ann arrived, and he allows the possibility that not every student who is not Bill arrived...' This correctly captures the compatibility of (91a) and (91c).

- (92) a. [*EXH*_{Alt} [ASSERT Every student except Dina arrived]]

Similarly, modal suspension readings are available for *besides*: while (93a) is incompatible with a statement that directly contradicts the exceptive inference in (93b), it is compatible with the same statement when embedded under *perhaps*. This demonstrates that the argument extends to *besides* as well.

- (93) a. Every student besides Dina arrived, ...
b. #and even she did!
c. and perhaps even she did!

In summary, our decision to separate the two steps – domain subtraction and negation of alternatives – in the syntax for the exceptive-additive *besides* is well-founded.

6 Extensions and challenges

6.1 *Besides* and collective predicates

So far, this paper has examined cases involving modified numerals with *besides* and distributive predicates within their scope. In this section, we will explore how numerals

modified by *besides* interact with collective predicates. We start by observing that *besides*-modified numerals are not compatible with collective predicates like *lift the piano together*. This is illustrated in (94).

- (94) a. *Ten students besides John lifted the piano together yesterday.
 b. *More than ten students besides John lifted the piano yesterday.
 c. *Exactly ten students besides John lifted the piano yesterday.
 d. *Fewer than ten students besides John lifted the piano yesterday.

We also observe a contrast between the cases above and those involving other collective predicates, such as *gather* and *meet*: these predicates are compatible with *besides* (at least to some extent).

- (95) a. Ten students besides John met downstairs.
 b. More than ten students besides John met downstairs.
 c. Exactly ten students besides John met downstairs.
 d. ?Fewer than ten students besides John met downstairs.
- (96) a. Ten students besides John gathered downstairs.
 b. More than ten students besides John gathered downstairs.
 c. Exactly ten students besides John gathered downstairs.
 d. ?Fewer than ten students besides John gathered downstairs.

One property that differentiates between the two types of predicates is their ability to license sub-entailments (i.e., inferences from groups to subgroups). For instance, (97a) entails (97b), but there is no entailment in either direction between (98a) and (98b). (98a) and (98b) describe two distinct events of piano lifting. Distributive predicates, of course, exhibit the same pattern as *gather* with respect to sub-entailments (also illustrated in (97)). It is reasonable to hypothesize that this property of the predicate within the scope of the numeral is ultimately responsible for whether a *besides*-construction will be acceptable with that numeral.

- (97) a. Ten students gathered downstairs/came.
 b. Nine students gathered downstairs/came.
- (98) a. Ten students lifted the piano together yesterday.
 b. Nine students lifted the piano together yesterday.

Our next step is the observation that the approach that we have developed so far does not capture this difference between the two types of predicates in their ability to co-occur felicitously with *besides*.

Let's consider the ungrammatical case with a downward-entailing numeral in (94d). If the maximality operator is inserted into the structure with *besides*, as shown in (99a), we obtain a well-formed meaning for this sentence. The resulting truth conditions are provided in (99b). The sentence is predicted to be true in two types of scenarios. In the first scenario, the largest group of students who lifted the piano without John consisted of fewer than 10 people. Additionally, there was a distinct event where a larger group of students, including John, lifted the piano together. In the second scenario, no group not including John lifted the piano, but there was a group with John that did the lifting. In this case, the degree that makes the existential quantification true is 0.

- (99) a. $[IP_6[\text{fewer than ten}][IP_5 2 [IP_4 \text{ EXH } [IP_3[\text{max } d_2] [IP_2 1 [IP_1 d_1 \text{ many students}$

- besides John lifted the piano yesterday]]]]]]
- b. $\llbracket (94d) \rrbracket^{g,w} = T$ iff $\exists d[d < 10 \ \& \ \max(\lambda n. \exists X[|X|=n \ \& \ X \text{ is a student}_w \ \& \ \neg(x \circ \text{John}) \ \& \ X \text{ lifted}_w \text{ the piano}])=d \ \& \ \max(\lambda m. \exists Y[|Y|=m \ \& \ Y \text{ is a student}_w \ \& \ Y \text{ lifted}_w \text{ the piano}]) \neq d]$

The challenge we are facing, thus, is that our proposal is not subtle enough to differentiate between predicates that license sub-entailments and the ones that do not.

It has been previously observed that the property of licensing sub-entailments is relevant in a sphere distinct from exceptives. Specifically, it has been reported that downward-entailing numerals behave differently depending on whether the predicate in the scope of the numeral is distributive or collective (Winter 2001, Ben-Avi and Winter 2003, Buccola 2015, Spector 2014, Buccola and Spector 2016).

The sentence in (100a), with the distributive main predicate *came*, exhibits the upper-bound reading discussed earlier: it means that there is no plurality of students who came with a cardinality larger than five. Additionally, this sentence can be true even if no students came at all.

- (100) a. Fewer than ten students came.
b. Fewer than ten students lifted the piano together yesterday.

The situation differs in (101b), where the collective reading of the main predicate *lifted the piano*, facilitated by *together*, is intended. Two properties distinguish this reading from those discussed in previous sections: First, it does not impose an upper-bound limit—the sentence is compatible with the possibility that a larger group of students (containing, say, twelve students) also lifted the piano; second, the sentence carries the existential inference that some students did lift the piano.

The idea we will explore here is that the solution proposed to account for the absence of upper boundedness in DE numerals with collective predicates like *lift the piano* in their scope also provides a solution for the interaction of collective predicates and *besides*. Specifically, we will consider an alternative theory of maximality based on logical strength (Buccola and Spector 2016, von Stechow et al. 2014, Dayal 1996, Rullmann 1995, Fox and Hackl 2006, Beck and Rullmann 1999, Schlenker 2012) and explore the predictions our approach to *besides* makes when it combines with this theory.

As Buccola and Spector (2016) show, applying this type of maximality operator generates non-upper-bound readings for DE numerals with collective predicates and upper-bound readings for such numerals with distributive predicates, as desired.

We will begin by defining the maximality operator and briefly outlining the independent motivation behind its introduction. Following Buccola and Spector (2016), we redefine our maximality function as follows:

- (101) For any property of numbers P , world w , and number n : $\max_{inf}(P)(w)(n) = 1$ iff n is P -maximal in w .
- (102) For any property of numbers P (i.e. function of type $\langle s \langle dt \rangle \rangle$), a number n is P -maximal in a world w iff one of the following two conditions holds:
- a. i. $P(w)(n) = 1$
ii. There is no m such that $P(w)(m)=1$ and $[\lambda v. P(v)(m)=1] \subsetneq [\lambda v. P(v)(n)=1]$
- b. i. There is no number k such that $P(w)(k) = 1$
ii. $n=0$

As with the definition of the maximality operator we have used before, a special amend-

ment to the maximality operator is required to account for the fact that distributive predicates with DE numerals are compatible with situations where no individual satisfying the predicate exists. The clause in (102b) covers this special case.

The clause in (102a) aims to capture the difference in upper-boundness of DE numerals depending on the properties of the predicate in their scope. While the proposition in (103a) entails the proposition in (103b), no entailment in any direction exists between (104a) and (104b), as they can only describe two independent lifting events.

- (103) a. $\lambda w. \exists X[|X|=10 \ \& \ X \text{ is a student}_w \ \& \ X \text{ came}_w]$
 b. $\lambda w. \exists X[|X|=9 \ \& \ X \text{ is a student}_w \ \& \ X \text{ came}_w]$
- (104) a. $\lambda w. \exists X[|X|=10 \ \& \ X \text{ is a student}_w \ \& \ X \text{ lifted}_w \text{ the piano together}]$
 b. $\lambda w. \exists X[|X|=9 \ \& \ X \text{ is a student}_w \ \& \ X \text{ lifted}_w \text{ the piano together}]$

Thus, the set of degrees picked by the function in (105a) in a world w will always be a singleton set; its only member will be simply the maximal number of comers in w . In contrast, the set picked by the function in (105b) might not be a singleton set. If, in w , two groups of students consisting of 9 and 10 people lifted the piano, the set picked by (105a) will contain two degrees, 9 and 10. This is because both of them are maximal under the definition of maximality given in (102).

- (105) a. $\lambda n. \max_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ came}_{w'}])(w)(n)$
 b. $\lambda n. \max_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ lifted}_{w'} \text{ the piano together}))(w)(n)$

Consequently, inserting this maximality operator in the scope of a downward entailing numeral ensures that the resulting interpretation is upper-bound if degree predicate licenses downward inferences and not upper-bound if the degree predicate does not do so.

To illustrate, the resulting truth conditions for our example with the distributive predicate in (106a) we predict that the sentence is false if two groups of students came: one consisting of 9 people and one consisting of 10. This is because the maximal number of comers is 19 and it is not smaller than 10. In contrast to this, the truth conditions for the sentence with the collective predicate predict that it is true in a parallel scenario where two groups consisting of 9 and 10 people lifted the piano. This is because under our new definition of maximality both 9 and 10 are the maximal numbers of piano-lifters.

- (106) a. $\llbracket (100a) \rrbracket^{g,w} = T \text{ iff } \exists n[n < 10 \ \& \ \max_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ came}_{w'}])(w)(n)]$
 b. $\llbracket (100b) \rrbracket^{g,w} = T \text{ iff } \exists n[n < 10 \ \& \ \max_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ lifted}_{w'} \text{ the piano together}))(w)(n)]$

The issue that this approach leaves unsolved is the existential entailment of sentences where downward-entailing numerals have collective predicates within their scope. As previously mentioned, to explain the lack of existential entailment with distributive predicates, a special clause must be added to the definition of the maximality operator. This clause ensures that if no plurality meets the existential quantification within the operator's scope, its application to the world and the predicate of degrees results in 0. In a scenario where no one lifted the piano yesterday, the truth conditions given in (106b) incorrectly predict the sentence to be true, contrary to the facts. For further

discussion of this issue, we refer readers to (Buccola and Spector 2016). Since this is a broader problem in the semantics of modified numerals and is not directly relevant to the interaction between numerals and *besides*, we leave it for future research.

What we will demonstrate below is that appealing to maximality based on informativity provides a natural explanation for the contrast between predicates that license sub-entailments and those that do not, in terms of their compatibility with *besides*. We maintain the same assumptions about LFs containing a modified numeral as before. For convenience, an LF for (107a) is provided in (107b). The only modification we introduce is in the denotation of the type shifter. MAX_{inf} now makes reference to the metalinguistic maximality function defined above. $[\text{MAX}_{inf} d_2]$ combines with its sister through intensional functional application.

- (107) a. Fewer than ten students came.
 b. $[\text{IP}_4 \text{fewer than ten } [\text{IP}_3 2 [\text{IP}_2 [\text{MAX}_{inf} d_2] [\text{IP}_1 1 d_1 \text{many students came}]]]]]$
 c. $[[\text{MAX}_{inf}]]^{g,w} = \lambda d. \lambda P_{\langle s, \langle dt \rangle \rangle}. \text{max}_{inf}(P)(w)(d)$

With these modifications to the notion of maximality, we retain the predictions regarding the interaction of *besides* with modified numerals that have distributive predicates in their scope from the previous version of the maximality operator. However, we now predict that the insertion of EXH will be vacuous if the predicate is collective and does not license sub-entailments. Collective predicates like *gather*, which do license sub-entailments, are predicted to pattern similarly to distributive predicates.

First, let's examine the LF in (108a) with a distributive predicate. The constituent denoted by IP_4 is the degree predicate (as shown in (109)); its degree argument will eventually be existentially bound by the numeral. This predicate selects a non-empty set precisely when adding John to the domain increases the number of comers.

In a scenario where nine students, including John, came overall there is a number smaller than 10—specifically 8—such that, in the domain that does not include John, 8 is the maximal number of comers. However, in the domain that includes John, 8 is not the maximal number of comers.

- (108) a. $[\text{IP}_5 \text{fewer than 10 } [\text{IP}_4 2 [\text{IP}_3 \text{EXH } [\text{IP}_2 [\text{MAX}_{inf} d_2] [\text{IP}_1 1 d_1 \text{many students besides John came}]]]]]$
 b. $[\text{IP}_5 \text{fewer than 10 } [\text{IP}_4 2 [\text{IP}_3 \text{EXH } [\text{IP}_2 [\text{MAX}_{inf} d_2] [\text{IP}_1 1 d_1 \text{many students besides John lifted the piano together}]]]]]$

- (109) $[[\text{IP}_4]]^{g,w} = \lambda n. \text{max}_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ \neg(x \circ \text{John}) \ \& \ X \text{ is a student}_{w'} \ \& \ \text{came}_{w'}])(w)(n)$
 $\ \& \ \neg \text{max}_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ came}_{w'}])(w)(n)$

Now let's consider the crucial case with a collective predicate that does not license sub-entailments in (108b). For the time being, let's set aside a possible situation where no group of students that does not include John lifted the piano. We cannot find a value for $g(2)$ (other than 0) that can make the prejacent of EXH with the meaning in (110a) and the negation of the only alternative distinct from the prejacent (given in (110b)) simultaneously true when applied to the same world.

- (110) a. $\lambda w. \text{max}_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ \neg(x \circ \text{John}) \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ lifted}_{w'} \text{ the piano}]) (w)(g(2))$
 b. $\lambda w. \neg \text{max}_{inf}(\lambda d. \lambda w'. \exists X[|X|=d \ \& \ X \text{ is a student}_{w'} \ \& \ X \text{ lifted}_{w'} \text{ the piano}]) (w)(g(2))$

Let's consider a possible value of 8 for $g(2)$ and a scenario where the prejacent is

true—where a group of students, not including John, consisting of 8 people, lifted the piano together. Given the upward-entailing properties of *many*, if there exists a group of 8 students, none of whom is John, who lifted the piano, it logically follows that there is a group of 8 students who lifted the piano.

Even if there is a larger group of students, consisting of 9 people (including John), who lifted the piano, both 8 and 9 would be considered the maximal number of students who lifted the piano according to the definition of maximality we’re using. Given that, if $g(2) = 8$, (110b) will be false when evaluated in the same world where (110a) is true. This holds for any other number (except 0, which we’ve set aside for now). Since the only alternative is entailed by the prejacent, EXH cannot negate it, and thus its application becomes vacuous. As a result, the LF in (108b) is predicted to be ruled out by the non-vacuity constraint.

Now let’s go back to the situation we previously set aside—where no plurality satisfies the existential quantification within the scope of the numeral, meaning that no group of students without John lifted the piano. In this case, the prejacent and the negated alternative can both be true if there is also at least one group of students including John who lifted the piano. Here, 0 becomes a possible value for $g(2)$ in both clauses in (109). Since the maximal number of students without John who lifted the piano is 0, (110a) is true. Because 0 is not the maximal number of lifters overall, (110b) is also true.

This scenario presents a challenge to the approach we’ve developed. However, as previously mentioned, this issue is more general and pertains to the semantics of numerals. Empirically, it is well established that sentences with DE numerals and collective predicates are false in situations where no plurality satisfying the predicate exists. Therefore, this scenario must be ruled out by a constraint that enforces the existential entailment for DE numerals with collective predicates in their scope. While the exact nature of this constraint remains unclear, it effectively rules out this scenario.

The use of informativity-based maximality was initially motivated by its application to downward-entailing numerals, but the incompatibility of numerals with *besides* and collective predicates that do not license sub-entailments is a more general phenomenon. This extends beyond downward-entailing numerals to include upward-entailing and non-monotonic ones, as discussed earlier in this section. This is directly predicted by the approach we have developed. As we have demonstrated, the issue arises at the level of the maximality operator. Therefore, the approach naturally extends to all other cases involving such collective predicates in (94).

To summarize, using the informativity-based version of the maximality operator, which has been independently argued for, enables us to account for the distinction between predicates that license sub-entailments and the ones that don’t in terms of their compatibility with modified numerals combined with *besides*.

6.2 *Besides* and upward scale predicates

We adopted a decomposition account of modified numerals, proposing that the positive inference contributed by *besides* in these contexts is computed below the scope of the numeral itself. Crucially, our analysis relied on the presence and availability of the maximality operator within the scope of the modified numeral. This ensured that the additive inference was computed on the non-monotonic constituent. Accordingly, we expect that in cases where the maximality operator cannot be inserted, modified numerals would not be compatible with *besides*.

It has been argued that the maximality operator cannot be inserted on top of upward-scale predicates such like *sufficient* in (111a) and certain constructions featuring existential modals, as seen in (111b). This is because there is no maximal degree that satisfies such predicates (Beck and Rullmann 1999, Fox and Hackl 2006, Buccola and Spector 2016). For example, if five people are sufficient, then any larger number of people would also be sufficient. Likewise, if five people can lift the piano, then it can also be lifted by a larger group of people (Buccola 2016).

- (111) a. Five people are sufficient to lift this piano.
b. Five people can lift this piano.

Even though (111a) and (111b) seemingly express the same meaning, we unexpectedly observe a difference in how the two constructions interact with *besides*: while *besides* can occur felicitously in modified numeral constructions involving *sufficient*, it is not acceptable in constructions that involve existential modals, as the contrast in (112) shows. The sentence in (112a) means something like this: ‘If we put John and any five other people together, the piano can be lifted.’ It is appropriate in a context where John has exceptional physical strength and is singled out for this reason. Our conjecture is that the difference lies in the syntax of the two constructions.

- (112) a. Five people besides John are sufficient to lift this piano.
b. *Five people besides John can lift this piano.

6.3 *Besides and sufficient*

Adapting the paraphrase proposed by Beck and Rullmann (1999) for similar sentences, we propose that (111a) can be paraphrased as shown in (113), notably with a predicate *available*. We will take seriously the idea that this predicate is covertly present in the structure of the sentence, a possibility briefly discussed in (Buccola and Spector 2016:p. 175). We will remain somewhat vague about the syntax of (111a), but we will assume that the constituent preceding *sufficient* contains a silent predicate *available*, as shown in the LF in (114).

- (113) It is possible (given the physical restriction of our world) that there are only five people available and the piano gets lifted

- (114) $[IP_4[IP_3[IP_2 \text{Five people available}] \text{are sufficient}] [IP_1 \text{to lift this piano}]]$

Instead of presenting the meaning of *sufficient* as containing existential quantification over possible worlds and a meaning component akin to *only*, as was proposed in (Beck and Rullmann 1999), we propose that it introduces universal and existential quantification over possible worlds and composes with two propositional arguments as shown in (115) (here R_{sym} stands for a similarity accessibility relation, R_{phys} represents identity in terms of physical properties).

- (115) $\llbracket \text{sufficient} \rrbracket^{g,w} = \lambda p. \lambda q. \forall w' [R_{sym}(w')(w) \ \& \ p(w') \rightarrow \exists w'' [R_{phys}(w'')(w') \ \& \ q(w'')]]$

The first propositional argument in (111a) is a proposition denoted by *five people are available*, and the second one is *to lift the piano*. The resulting truth conditions for (111a) given in (116) capture the idea behind the paraphrase in (115): in every world where five people are available (even in the ones where only five people are available), there is a possibility that the piano gets lifted.

$$(116) \quad \llbracket (111a) \rrbracket^{g,w} = T \text{ iff } \forall w' [R_{sym}(w')(w) \& \text{ five people are available in } w' \rightarrow \exists w'' [R_{phys}(w'')(w') \& \text{ the piano gets lifted in } w'']]$$

Given that under these assumptions *besides* in (112a) occurs inside one of the propositional arguments of *sufficient*, we can compute the additive inference contributed by *besides* inside this clause, following the steps laid out in this paper. The LF for (112a) that we will assume is shown in (117).

$$(117) \quad [IP_7 [IP_6 \text{ five } [IP_5 2 [IP_4 [\max_{inf} d_2] [IP_3 1 [IP_2 d_1 \text{ many people besides John available}]]]] \text{ are sufficient}] [IP_1 \text{ to lift this piano}]]$$

The predicted truth conditions are as follows: in every world where there is a number equal to 5 or larger, such that this number is the maximal available people excluding John, but it is not the maximal number of available people, the piano gets lifted.

$$(118) \quad \llbracket (117) \rrbracket^{g,w} = T \text{ iff } \forall w' [R_{sym}(w')(w) \& \exists n [n \geq 5 \& \max_{inf} (\lambda w'' . \lambda d . \exists X [|X|=d \& \neg (X \circ \text{John}) \& X \text{ is a person}_{w''} \& X \text{ is available}_{w''})(w')(n) \& \neg \max_{inf} (\lambda w'' . \lambda d . \exists X [|X|=d \& X \text{ is a person}_{w''} \& X \text{ is available}_{w''})(w')(n) \rightarrow \exists w'' [R_{phys}(w'')(w') \& \text{ the piano gets lifted in } w'']]]$$

This accurately captures the truth conditions of the sentence, including the additive inference. Essentially, with these syntactic assumptions, when *besides* appears in a sentence with *sufficient*, it actually occurs inside a clause of a familiar form, specifically one with a distributive predicate in the scope of the modified numeral. As a result, this case can be reduced to those we previously analyzed, allowing us to apply the same interpretative mechanisms.

6.4 Besides and can

We conjecture that the sentence in (111b), in contrast, does not have a bi-clausal structure. Instead, it has a structure as shown in (119), where the subject is reconstructed below the existential modal.

$$(119) \quad [IP_5 \text{ can } [IP_4 \text{ five } [IP_3 2 [IP_2 [\max_{inf} d_2] [IP_1 1 d_1 \text{ many people lifted the piano together}]]]]]$$

The resulting interpretation is as shown in (120).

$$(120) \quad \llbracket (119) \rrbracket^{g,w} = T \text{ iff } \exists w' [R_{phys}(w')(w) \& \max_{inf} (\lambda d . \lambda w'' . \exists X [|X|=d \& X \text{ is a person}_{w''} \& X \text{ lifted the piano}_{w''}])(w')(5)]$$

Given the LF in (119), to explain the ungrammaticality of *besides* in this construction, we only need to observe that we have already established that there is no place in the preadjacent of the modal (IP_4) where the EXH associated with *besides* can be inserted non-vacuously. This explains why *besides* cannot occur felicitously in this construction.

7 Besides in wh-questions

Besides also occurs with wh-phrases. In that case, too, it contributes additive inferences.

$$(121) \quad \text{a. } \textit{Who besides Ann passed?} \quad \rightsquigarrow \text{Ann passed}$$

- b. Which student(s) besides Ann passed? $\rightsquigarrow$ Ann passed

Just like in the cases considered so far, *besides* should contribute subtraction, this time in the restrictor of the wh-word. Now, the wh-phrase scopes over the question operator $?_{op}$, as standardly assumed. Since $?_{op}$ returns a question denotation, the wh-phrase itself is composed with such a denotation. The consequence of this is that *EXH*, which is necessary to generate the additive inference, will also have to apply to a question denotation. Assuming though that question-denotations, at least at the level relevant here, denote the true answer to the question, *EXH* could apply to propositions even in the case of (121). This would, however, raise another issue. A true answer by definition cannot be negated without incurring a contradiction. So *EXH* would be necessarily vacuous making both of (121a) and (121b) ungrammatical. In the following, we suggest that *EXH* applies to the intension of the weakly exhaustive answer of the question. Unlike strongly exhaustive answers, such answers to wh-questions with *besides* stand in an entailment relation with their alternatives. Before showing this, an alternative analysis that might come to mind is shown to be untenable.

7.1 An ill-fated approach: applying *EXH* in the question nucleus

On the copy-theory of movement traces correspond to definite descriptions with a full copy of the NP inside the trace (e.g. Fox 2002, Sauerland 2004). One might thus ask whether *besides* might not simply contribute subtraction inside the trace. Assume for concreteness the the LF (122) for (121b). Here *besides Ann* is not part of the higher copy – or alternatively left uninterpreted there – and *EXH* occurs under $?_{op}$. At first blush this appears to allow for the possibility that nothing special might have to be said about *besides* in wh-questions at all.

$$(122) \quad [[\text{which student}(s)] \lambda_2 [?_{op} [EXH_{Alt} [[\text{the}_2 \text{ student}(s) \text{ besides Ann }] \text{ passed }]]]]$$

Leaving certain details aside, the trace ultimately must contribute the individual identical to the assignment to index 2 on the definite article. This means that the content of *student(s) besides Ann* only figures in the presuppositional component:

$$(123) \quad \llbracket \text{the}_2 [\text{student}(s) \text{ besides Ann }] \rrbracket^{g,w} = g(2) \\ \text{defined only if } g(2) \in \llbracket \text{student}(s) \text{ besides Ann } \rrbracket^w$$

Assume the students are Ann, Betty, Carl, and Dan. In that case, the alternatives for *EXH* look as follows in the singular case. Since for the prejacent to be true, $g(2)$ must be one of Betty, Carl, and Dan, the same must be the case in the alternative where Ann is put back into the set of students. Consequently, the two alternatives are equivalent whenever the prejacent has a defined truth-value and *EXH* cannot exclude anything. Regardless of the violation of the no-vacuity constraint, no additive inference is thus generated.

$$(124) \quad \left\{ \begin{array}{l} \lambda w.g(2) \text{ passed in } w, \text{ defined only if } g(2) \in \{\text{Betty, Dan, Carl}\} \\ \lambda w.g(2) \text{ passed in } w, \text{ defined only if } g(2) \in \{\text{Ann, Betty, Dan, Carl}\} \end{array} \right\}$$

It is easy to see that this result carries over to the plural case. This is so because the set of individuals in the prejacent – atomic or non-atomic – is a proper subset of the one in the alternative.¹⁵

¹⁵In light of this one might entertain the idea that *student(s) besides Ann* contributes to the assertive

7.2 Applying EXH at the answer-level

7.2.1 Weakly exhaustive answers

The weak exhaustive answer to (125) is the proposition entailing for everyone who passed that they did so. The usefulness of this notion has been demonstrated in the literature on embedded questions (cf. Heim 1994, Beck and Rullmann 1999, Klinedinst and Rothschild 2011 a.m.o.).

(125) *Who passed?*

For what follows the precise composition of wh-questions like (125) will not be really crucial. For concreteness we will assume wh-questions denote sets of mention-some answers, as proposed by Karttunen (1977) from which the weakly exhaustive answer can be straightforwardly derived.¹⁶ For concreteness, we assume that the LFs for questions like (121a) and (125) are as in (126a) and (126b), respectively.¹⁷

- (126) a. $[ANS_1 [[wh [-o]] \lambda_2 [?_{op} [t_2 \text{ passed}]]]]$
 b. $[ANS_1 [[wh [-o \text{ besides Ann}]] \lambda_2 [?_{op} [t_2 \text{ passed}]]]]$

Notice that the wh-phrases take scope over $?_{op}$, as mentioned above and there is abstraction over their traces. The lexical entries for *wh* and $?_{op}$ could then be as follows, assuming that *-o* in (126) contributes the (characteristic function of the) set of humans. Notice also that (127a) returns functions from worlds to sets of propositions. We call this the K(arttunen)-intension as opposed to the K-extension, obtained from applying the K-intension to a world.

- (127) a. $\llbracket wh- \rrbracket = \lambda f_{et} \cdot \lambda g_{\langle e, \langle st, t \rangle \rangle} \cdot \lambda w_s \cdot \lambda p_{st} \cdot \exists x [f(x)(w) = 1 \wedge g(x)(p) = 1]$
 b. $\llbracket ?_{op} \rrbracket = \lambda p_{st} \cdot \lambda q_{st} \cdot q = p$

For (125), this yields the K-extension in w^* given in (128), assuming the four individuals in w^* are as given. It becomes clear that the wh-phrase contributes the variation in individuals considered among the mention-some answers.

- (128) $\llbracket [[wh [-o]] \lambda_2 [?_{op} [t_2 \text{ passed}]]] \rrbracket^g(w^*)$
 $= \{p_{st} : \exists x [x \text{ is human in } w^* \wedge p = \lambda w'. x \text{ passed in } w']\}$

component as well. This is independently questionable, as mention-some answers – the propositions in the denotations of wh-questions (see below) – at least on the surface do not appear to involve definite descriptions like *the student(s)*. But even ignoring this non-trivial issue, such an account would face the problem that *besides* in singular definite descriptions is ungrammatical and leads to exceptive rather than additive inferences to the extent that it is acceptable in plural definite descriptions:

- (i) a. **The student besides Ann passed.*
 b. *The students besides Ann (all) passed.* $\rightsquigarrow$ Ann didn't pass

¹⁶In principle, it would be possible for such sets to be downward closed as assumed in inquisitive semantics (e.g. Ciardelli et al. 2019). The relevant notion of entailment at the level of the weakly exhaustive answer could be obtained here as well. For simplicity, we stick with the less complex denotations.

¹⁷We revert to LFs without articulated copies. Assuming the copy-theory of movement would be compatible with what will be said below, however. Given the discussion above, EXH must not occur under $?_{op}$. EXH can, however, apply to the whole question as at this level the denotation is essentially the same as on the non-articulated LFs in the text.

$$= \left\{ \begin{array}{l} \lambda w. \text{Ann passed in } w \\ \lambda w. \text{Betty passed in } w \\ \lambda w. \text{Carl passed in } w \\ \lambda w. \text{Dan passed in } w \end{array} \right\}$$

Given our assumptions regarding *besides* the K-extension in $g(w_3)$ for (121a) therefore looks as in (129).

$$(129) \quad \begin{aligned} & \llbracket \llbracket wh \llbracket -o \text{ besides Ann } \rrbracket \lambda_2 \llbracket ?_{op} \llbracket t_2 \text{ passed } \rrbracket \rrbracket \rrbracket^{g(w^*)} \\ &= \{p_{st} : \exists x [x \text{ is human in } w^* \wedge x \text{ is not Ann} \wedge p = \lambda w'. x \text{ passed in } w']\} \\ &= \left\{ \begin{array}{l} \lambda w. \text{Betty passed in } w \\ \lambda w. \text{Carl passed in } w \\ \lambda w. \text{Dan passed in } w \end{array} \right\} \end{aligned}$$

ANS_1 in (126a) and (126b) above turns the K-intensions into their weakly exhaustive answers. This is defined in (130), following Heim (1994). It takes a world w and a K-intension Q and returns the conjunction of the mention-some answers in the K-extension of Q in w .

$$(130) \quad \llbracket ANS_1 \rrbracket = \lambda Q_{\langle s, \langle st, t \rangle \rangle}. \lambda w_s. \bigwedge \{p : p \in Q(w) \wedge p(w) = 1\}$$

If in world w^* Ann and Bill passed and Carl and Dan did not, there are two true mention-some answers in (128) and only one in (129). The weakly exhaustive answers to (125) and (121a) are thus as in (131a) and (131b), respectively. The former asymmetrically entails the later.

$$(131) \quad \begin{aligned} \text{a. } & \llbracket ANS_1 \rrbracket (\llbracket \llbracket wh \llbracket -o \rrbracket \lambda_2 \llbracket ?_{op} \llbracket t_2 \text{ passed } \rrbracket \rrbracket \rrbracket^{g,w})(w^*) \\ &= \lambda w. \text{Ann and Bill passed in } w \\ \text{b. } & \llbracket ANS_1 \rrbracket (\llbracket \llbracket wh \llbracket -o \text{ besides Ann } \rrbracket \lambda_2 \llbracket ?_{op} \llbracket t_2 \text{ passed } \rrbracket \rrbracket \rrbracket^{g,w})(w^*) \\ &= \lambda w. \text{Bill passed in } w \end{aligned}$$

In a world in which only Bill passed, the weakly exhaustive answers are equivalent. It is thus easy to see that whatever is the case in the world of evaluation, the weakly exhaustive answer to (125) in that world entails the one to (121a) in that same world.¹⁸ We can thus generalize entailment from propositions as in (132a) to intensions thereof as in (132):

$$(132) \quad \begin{aligned} \text{a. } & \text{If } p, q \in D_{st}, p \text{ entails } q, p \Rightarrow q, \text{ iff for any world } w \text{ such that } p(w) = 1, \\ & q(w) = 1.^{19} \\ \text{b. } & \text{If } p, q \in D_{\langle s, st \rangle}, p \Rightarrow q \text{ iff for any worlds } w, w' \text{ such that } p(w)(w') = 1, \\ & q(w)(w') = 1. \end{aligned}$$

7.2.2 Applying EXH to the extension of a weakly exhaustive answer

What would the application of EXH to (131b) – the weakly exhaustive answer to the wh-question *Who besides Ann passed?* in (121a) – yield? This depends on what the set

¹⁸In a world where no one passed, the weakly exhaustive answers to both questions amount to the tautology. Assuming that wh-questions, all else being equal, come with an existential presupposition that one mention-some answer in their K-denotation must be true (e.g. Dayal 1996), this defective case is avoided. See the discussion in section 7.3.1 below, where such an existential presupposition is encoded as part of a uniqueness presupposition.

¹⁹In the propositional case $\subseteq$ and $\Rightarrow$ are used interchangeably.

of alternatives to (131b) would be. If the set of alternatives to (131b) Alt_{str} included simply the stronger (131a), as one might expect given the general approach to *besides* in this paper, we would not get the desired result. First, it does not make much sense conceptually to negate (131a) given that it is itself a weakly exhaustive answer and is thus by definition true. Second, if EXH were to negate it nonetheless, (133) would obtain. Regardless of its falsity in w^* , (133) is not the desired result, as an exceptive inference rather than an additive one would result for (121a).²⁰

$$(133) \quad \llbracket EXH_{Alt_{str}} \rrbracket^g(\llbracket ANS_1 \rrbracket(\llbracket Who besides Ann passed? \rrbracket^{g,w})(w^*)) \\ = \lambda w. \text{Bill passed in } w \wedge \text{Ann did not pass in } w$$

Should we change our assumptions regarding alternatives for wh-questions with *besides*? One possibility coming to mind is to take the set of alternatives Alt'_{str} to contain the weakly exhaustive answers to (131b) in alternative worlds.²¹ If EXH were to strengthen (131b) with all such non-entailed propositions, the proposition in (134) would result, but (134) has the same issues as (133).²²

$$(134) \quad \llbracket EXH_{Alt'_{str}} \rrbracket^g(\llbracket ANS_1 \rrbracket(\llbracket Who besides Ann passed? \rrbracket^{g,w})(w^*)) \\ = \lambda w. \text{Bill passed in } w \wedge \text{Ann and Carl and Dan did not pass in } w$$

7.2.3 Applying EXH to the intension of a weakly exhaustive answer

For these reasons we suggest that EXH should apply to the intension of the weakly exhaustive answer to the wh-question with *besides*. For (121a) the representation should look as follows.

$$(135) \quad [EXH_{Alt_{str}}^q [ANS_1 [who besides Ann passed]]]$$

The idea now is that this question-level EXH^q returns a strengthened intension of a weakly exhaustive answer, which means that EXH^q also has an argument slot for a world. But it also follows from this that the definition of EXH^q must be amended when compared to the operator used in the propositional case so far. Consider (136). This function takes a world w and the intension of proposition p and returns true just in case the extension of p in w is true in w and any alternative q not entailed by p has an extension in w different from the one of p in w .^{23 24 25}

²⁰From here on the internal structure of the wh-question is not given in detail whenever it is not necessary.

²¹We would need to obtain alternatives by varying the world in which the answers are considered. This could be done, for instance, by representing worlds in the representation.

²²If w^* where a world where only Bill passed, this would be the strongly exhaustive answer to the question. This answerhood notion is useful, as it is necessary to derive, for instance, the correct inferences for wh-questions embedded under *know* (Groenendijk and Stokhof 1984).

²³For now we refer to this operator as EXH^q , assuming it is applicable to only questions. In section 7.4.3 we return to the issue of how to generalize this operator to the propositional case.

²⁴It would also be possible to define EXH^q as in (i). The difference here is that the non-entailment between preajacent and alternatives is based on the extension in w rather than the intension. As discussed for the weakly exhaustive answer in (131) in section 7.2.1, this would not make a difference for our cases.

(i) $\llbracket EXH_{Alt_{str}}^q \rrbracket = \lambda w_s. \lambda p_{\langle s, st \rangle}. p(w)(w) = 1 \wedge \forall q \in Alt_{str} [p(w) \not\Rightarrow q(w) \rightarrow p(w) \neq q(w)]$

²⁵For the question case we adopt a formalization without innocent exclusion, as this option is more perspicuous. Given that in the case of *besides* there is only one alternative different from the preajacent, this does not seem to have big consequences for the present cases. Adopting innocent exclusion for questions

$$(136) \quad \llbracket EXH_{Alt_{str}}^q \rrbracket = \lambda p_{\langle s, st \rangle} . \lambda w_s . p(w)(w) = 1 \wedge \forall q \in Alt_{str} [p \not\Rightarrow q \rightarrow p(w) \neq q(w)]$$

For EXH^q in (135), Alt_{str} is as in (137). The members of this set are intensions of weakly exhaustive answers to alternative wh-questions, namely to *Who besides Ann passed?* and to *Who passed?*. This is in line with the general assumptions in this paper.

$$(137) \quad Alt_{str} = \left\{ \begin{array}{l} [ANS_1 [who\ besides\ Ann\ passed]] \\ [ANS_1 [who\ passed]] \end{array} \right\}$$

Remember that the second alternative in (137) entails the first one. Given this (135) returns the (characteristic function of the) set of worlds w in (138) where the extensions of the weakly exhaustive answers to *Who besides Ann?* and *Who passed?* differ and where the extension of the former is true. As discussed in (131) above, due to the fact that the K-extension of the former question is necessarily a proper subset of the K-extension of the latter, the only way for the extensions of the weakly exhaustive answers in w to differ is for w to be a world where Ann passed. (138) therefore derives the desired additive inference of (121a).²⁶

$$(138) \quad \begin{aligned} & \llbracket [EXH_{Alt_{str}}^q [ANS_1 [who\ besides\ Ann\ passed]]] \rrbracket^{g,w} \\ &= \lambda w . [\llbracket [ANS_1] \rrbracket (\llbracket [who\ besides\ Ann\ passed] \rrbracket^{g,w})(w)(w) = 1 \wedge \\ & \quad \forall q \in Alt [\lambda w' . \llbracket [ANS_1] \rrbracket (\llbracket [who\ besides\ Ann\ passed] \rrbracket^{g,w})(w') \not\Rightarrow q \rightarrow \\ & \quad \llbracket [ANS_1] \rrbracket (\llbracket [who\ besides\ Ann\ passed] \rrbracket^{g,w}(w) \neq q(w))] \\ &= \lambda w . [\llbracket [ANS_1] \rrbracket (\llbracket [who\ besides\ Ann\ passed] \rrbracket^{g,w})(w)(w) = 1 \wedge \\ & \quad \llbracket [ANS_1] \rrbracket (\llbracket [who\ besides\ Ann\ passed] \rrbracket^{g,w}(w) \neq \llbracket [ANS_1] \rrbracket (\llbracket [who\ passed] \rrbracket^{g,w}(w))] \end{aligned}$$

7.3 Maximal true answer

7.3.1 Singular *which*-questions

Returning now to the singular *which*-question in (121b) repeated in (139b), we notice that just like the regular singular *which*-question in (139a) it comes with a uniqueness inference. The obligatoriness of that inference is brought out clearly in the discourse in (140). Reply A is odd because it contradicts the uniqueness inference of question Q.

- (139) a. *Which student passed?* $\rightsquigarrow$ exactly one student passed
b. *Which student besides Ann passed?* $\rightsquigarrow$ exactly one student besides Ann passed

- (140) Q: *Which student (besides Ann) passed?*
A: *#Betty and Carl.*

could look as follows: two weakly exhaustive answers q and r are symmetric to each other given weakly exhaustive answer p only if making p and q differ entails making p and r the same and making p and r differ entails making p and q the same. q and r are then innocently excludable given p just in case q and r are not symmetric to each other given p .

²⁶Remember we are assuming that wh-questions have an existential presupposition that one of the mention-some answers be true. Notice, however, that even without this presupposition (138) would be non-tautologous if Ann passed and no one else did. In that case, the weakly exhaustive answer to the first question would be the tautology but the one to the former would be the proposition that Ann passed. That is, the additive inference is derived regardless of the assumed existential presupposition. We might want to rule out such cases nevertheless on the basis that an embedded constituent would not have a defined truth-value. Note, however, that in a Strong-Kleene or a supervaluations approach to presupposition projection one would not need to make this assumption (e.g. Fox 2008, 2013).

Following Dayal (1996), we can bring this inference about by adding a uniqueness presupposition to ANS_1 as in (141). (141) requires that there be a true mention-some answer in the K-extension of question Q in w that entails all other true mention-some answers in it. That is, it requires the existence of a maximal true mention-some answer.

$$(141) \quad \llbracket ANS_1 \rrbracket = \lambda Q_{\langle s, \langle st, t \rangle \rangle} . \lambda w_s : \\ \exists p[p \in Q(w) \wedge p(w) = 1 \wedge \forall q[q \in Q(w) \wedge q(w) = 1 \rightarrow p \subseteq q]] . \\ \wedge \{p : p \in Q(w) \wedge p(w) = 1\}$$

Assuming Ann, Betty, Carl, and Dan are the students in w^* , the K-extensions of (139a) and (139b) look as in (142a) and (142b), respectively. Now, neither of the mention-some answers in either of the two sets entails another mention some answer. Consequently, the uniqueness requirement of ANS_1 in (141) can only be satisfied if exactly one the propositions is true, i.e., if exactly one of the students passed.

$$(142) \quad \begin{array}{ll} \text{a.} & \llbracket \text{Which student passed?} \rrbracket^{g,w}(w^*) = \left\{ \begin{array}{l} \lambda w. \text{Ann passed in } w \\ \lambda w. \text{Betty passed in } w \\ \lambda w. \text{Carl passed in } w \\ \lambda w. \text{Dan passed in } w \end{array} \right\} \\ \text{b.} & \llbracket \text{Which student besides Ann passed?} \rrbracket^{g,w}(w^*) = \left\{ \begin{array}{l} \lambda w. \text{Betty passed in } w \\ \lambda w. \text{Carl passed in } w \\ \lambda w. \text{Dan passed in } w \end{array} \right\} \end{array}$$

How does this interact with our treatment of *besides* in wh-questions laid out in the preceding section 7.2? Applying EXH^q to the K-extension of (142b) in world w^* returns the (characteristic function of the) set of worlds in which that K-extension is true and where moreover this extension differs from the K-extension of (142a). Assume a world w' where only Betty passed. Then the K-extensions are equivalent, and w' is thus not in that set of worlds. Consider now a world w'' where Betty and Carl passed. Here the uniqueness requirement is not satisfied in either, and w'' is again not in the set of worlds. Finally consider a world w''' where Betty and Ann passed. Here the uniqueness requirement of ANS_1 is satisfied in the case of (142b) but not in the case of (142a). That is, the extensions of the weakly exhaustive answers in w''' to (142b) and (142a) differ. The former has a defined value but the latter does not. World w''' is in the set of worlds returned by application of EXH^q therefore. This means that the additive inference of (139b) is derived.

7.3.2 Closure under plural-formation

Wh-questions other than singular *which*-questions do not come with a uniqueness requirement. Given that we assumed in section 7.2 K-extensions for *Who passed?* and *Who besides Ann passed?* parallel to those in (142a) and (142b), respectively, uniqueness effects would, however, be wrongly predicted by the modification of ANS_1 in (141). Before turning to these examples consider, however, the plural *which*-questions in (143a) and (143b) first. These also do not give rise to uniqueness inferences. Thus their K-extensions should be different from those in (141a) and (141b) as well.

$$(143) \quad \begin{array}{ll} \text{a.} & \text{Which students passed?} \\ \text{b.} & \text{Which students besides Ann passed?} \end{array} \quad \rightsquigarrow \text{Ann passed}$$

And indeed, it is clear what makes their K-extensions different. The reason is the

plural-marking on *student* necessitating application of the *-operator. The entry for the *-operator is repeated in (144) from above.

$$(144) \quad \llbracket * \rrbracket = \lambda f_{et}. \lambda x_e. \forall y [y \leq_{at} x \rightarrow f(y) = 1]$$

For simplicity, the LFs for the two questions in (143) are assumed to be as in (145). Crucially, * applies after subtraction by *besides* has been carried out. That is, it pluralizes the whole of *student besides Ann*. This has the consequence that Ann will neither be an element of the restrictor set directly nor be a part of a plurality in it.²⁷

$$(145) \quad \begin{array}{ll} \text{a.} & \llbracket [\textit{which} \textit{*student}] \lambda_2 [?_{op} [t_2 \textit{passed}]] \rrbracket \\ \text{b.} & \llbracket [\textit{which} \textit{*[student besides Ann]}] \lambda_2 [?_{op} [t_2 \textit{passed}]] \rrbracket \end{array}$$

The resulting K-extensions of (143a) and (143b) are therefore as in (146a) and (146b), respectively.

$$(146) \quad \begin{array}{ll} \text{a.} & \llbracket [\textit{Which students passed?}] \rrbracket^{g,w}(w^*) = \left\{ \begin{array}{l} \lambda w. \textit{Ann passed in } w \\ \lambda w. \textit{Betty passed in } w \\ \dots \\ \lambda w. \textit{Ann} \oplus \textit{Betty passed in } w \\ \dots \\ \lambda w. \textit{Ann} \oplus \textit{Betty} \oplus \textit{Carl} \oplus \textit{Dan passed in } w \end{array} \right\} \\ \text{b.} & \llbracket [\textit{Which students besides Ann passed?}] \rrbracket^{g,w}(w^*) = \left\{ \begin{array}{l} \lambda w. \textit{Betty passed in } w \\ \dots \\ \lambda w. \textit{Betty} \oplus \textit{Carl} \oplus \textit{Dan passed in } w \end{array} \right\} \end{array}$$

There are now mention-some answers about pluralities in the denotations. Crucially, these are partially ordered by entailment. For instance, if Betty and Ann passed in w , the maximal true mention-some answer in (146a) is the proposition that the plurality $\textit{Ann} \oplus \textit{Betty}$ passed, which entails the propositions that Ann passed and that Betty passed.²⁸ This means that in order to satisfy the uniqueness requirement of $\textit{ANS}_1$ it is not necessary anymore for exactly one mention-some answer to be true. Now, clearly in a world w where both Ann and Betty passed the weakly exhaustive answers to (146a) and (146b) differ. w would thus be in the set returned by the exhaustification of the weakly exhaustive answer to (146b). It is easy to see that similar to singular *which*-questions, a world w' where Ann did not pass makes the weakly exhaustive answers equivalent, whereby w' would not be in that strengthened answer. That is, the additive inference that Ann passed is again derived.

It is also clear that this solution can be carried over to *Who passed?* and *Who besides Ann passed?*. All we need to assume is that the restrictor of the *wh*-expression is inherently pluralized. Then again no uniqueness inference will be derived, but the additive inference would still be obtained.

²⁷This assumption is not crucial at all. One could also apply pluralization to *student* directly and redefine subtraction in such a way that it applies pointwise to each entity in that pluralized predicate.

²⁸This, of course, assumes that in addition to what is said in the text distributivity is contributed by the verbal predicate, possibly by an operator. Thereby the plurality $\textit{Ann} \oplus \textit{Betty}$ passing entails that both Ann and Betty passed.

7.3.3 Strongly exhaustive answers

All else being equal, a wh-question asks for the strongly exhaustive answer rather than the weakly exhaustive answer. While usually motivated on the basis of embedded wh-questions, this can also be seen with matrix wh-questions. For instance, B's reply to A's question in (147) suggests that only Ann passed. Now, if the answerhood condition to a wh-question is based on its semantic content – a standard assumption in the literature – then B's reply should express the weakly exhaustive answer given what has been said so far. From B's additional inferences would not follow. It appears that something stronger than the weakly exhaustive answer is needed.

- (147) A: *Who passed?*
 B: *Ann passed.* $\rightsquigarrow$ Bill and Carl did not pass

The strongly exhaustive answer to a question states that the weakly exhaustive answer is what it is in the actual world (Groenendijk and Stokhof 1984). That is, in addition to entailing all the true propositions in the K-denotation it also entails the negation of all the false ones. This could be implemented by Heim's 1994 answer operator Ans_2 in (148).

$$(148) \quad \llbracket Ans_2 \rrbracket = \lambda Q_{\langle s, \langle st, t \rangle \rangle}. \lambda w_s. \lambda w'_s. \bigcap \{p : p \in Q(w) \wedge p(w) = 1\} = \\ \bigcap \{p : p \in Q(w') \wedge p(w') = 1\}$$

Consider what would happen if wh-questions with *besides* involved Ans_2 . Assuming the four individuals from above, in a world w where only Ann and Bill passed, the strongly exhaustive answer to *Who besides Ann passed?* and its alternative differ:

- (149) a. $\llbracket Ans_2 \rrbracket(\llbracket \text{Who besides Ann passed?} \rrbracket^{g,w})(w)$
 $= \lambda w. \text{Bill passed in } w \text{ and Carl and Dan did not pass in } w$
 b. $\llbracket Ans_2 \rrbracket(\llbracket \text{Who passed?} \rrbracket^{g,w})(w)$
 $= \lambda w. \text{Ann and Bill passed in } w \text{ and Carl and Dan did not pass in } w$

Moreover, if only Bill passed in w , the strongly exhaustive answers would also differ:

- (150) a. $\llbracket Ans_2 \rrbracket(\llbracket \text{Who besides Ann passed?} \rrbracket^{g,w})(w)$
 $= \lambda w. \text{Bill passed in } w \text{ and Carl and Dan did not pass in } w$
 b. $\llbracket Ans_2 \rrbracket(\llbracket \text{Who passed?} \rrbracket^{g,w})(w)$
 $= \lambda w. \text{Bill passed in } w \text{ and Ann and Carl and Dan did not pass in } w$

It becomes clear that regardless of what the world is like, the strongly exhaustive answers to the two questions always differ. All else being equal, EXH^q applied to (150a) would thus be vacuous, which would violate the non-vacuity constraint. Crucially, the additive inference would not be derived for the *besides*-question.

Notice now that B's reply in (151) to A's *besides*-question still suggests strong exhaustivity. This would be problematic if the additive inference were dependent on Ans_1 together with EXH^q and the strong exhaustivity inference were dependent on Ans_2 .

- (151) A: *Who besides Ann passed?* $\rightsquigarrow$ Ann passed
 B: *Bill passed.* $\rightsquigarrow$ Carl did not pass

Fortunately, the strongly exhaustive answer can be derived from Ans_1 via another EXH

(see Klinedinst and Rothschild 2011, Uegaki 2015, Mayr 2019 a.o.). In a nutshell, the alternatives for *EXH* will be the potential weakly exhaustive answers derived by variation of the world argument of Ans_1 . Assume for illustration that only Ann and Bill passed in w . Then the weakly exhaustive answer to *Who besides Ann passed?* in w says that Ann and Bill passed. For the strongly exhaustive answer all alternative potential weakly exhaustive answers entailing that Carl and Dan passed will be negated. Arguments for this kind of treatment come from observations about the inferences licensed by embedding of wh-questions under predicates like *predict* (cf. Heim 1994, Klinedinst and Rothschild 2011 a.o.). Arguably such data necessitate a notion of answerhood between the weakly and the strongly exhaustive one in terms of strength that is inexpressible directly via an operator but expressible by strengthening the weakly exhaustive answer.

7.4 Imposing asymmetric entailment

7.4.1 Wh-questions with *but*

Wh-questions with *but* like (152) are ungrammatical as information-seeking questions. It turns out that if we assume the Gajewski/Hirsch-alternatives for *but*, the treatment of wh-questions with *besides* just outlined predicts (152) to be grammatical on such a reading.

(152) **Who but Ann passed?*

Assume there are three individuals Ann, Betty, and Carl. On the Gajewski/Hirsch-analysis, the alternatives used by EXH^q in the strengthening of the weakly exhaustive answer to (152) are as in (153). These are intensions of weakly exhaustive answers to wh-questions with *but* differing in which individual gets subtracted.

$$(153) \quad Alt_{str} = \left\{ \begin{array}{l} [ANS_1 [who \text{ but } Ann \text{ passed}]] \\ [ANS_1 [who \text{ but } Betty \text{ passed}]] \\ [ANS_1 [who \text{ but } Carl \text{ passed}]] \end{array} \right\}$$

In a world w with the three individuals given, the K-extensions of the wh-questions embedded under ANS_1 in (153) look as follows. Notice that here the pluralization assumption for the restrictor of *who* just discussed is adopted:

$$(154) \quad \begin{array}{ll} \text{a.} & \llbracket Who \text{ but } Ann \text{ passed} \rrbracket^{g,w}(w) = \left\{ \begin{array}{l} \lambda w. Betty \text{ passed in } w \\ \lambda w. Carl \text{ passed in } w \\ \lambda w. Betty \oplus Carl \text{ passed in } w \end{array} \right\} \\ \text{b.} & \llbracket Who \text{ but } Betty \text{ passed} \rrbracket^{g,w}(w) = \left\{ \begin{array}{l} \lambda w. Ann \text{ passed in } w \\ \lambda w. Carl \text{ passed in } w \\ \lambda w. Ann \oplus Carl \text{ passed in } w \end{array} \right\} \\ \text{c.} & \llbracket Who \text{ but } Carl \text{ passed} \rrbracket^{g,w}(w) = \left\{ \begin{array}{l} \lambda w. Ann \text{ passed in } w \\ \lambda w. Betty \text{ passed in } w \\ \lambda w. Ann \oplus Betty \text{ passed in } w \end{array} \right\} \end{array}$$

When does the weakly exhaustive answer to (154a) differ from the weakly exhaustive answer to both (154b) and (154c)? There are two options. Option (i) are worlds where at least Ann passed. In that case the weakly exhaustive answers to (154b) and (154c) entail that Ann passed, which the one to (154a) can never do given the absence of a

mention-some answer about Ann from (154a). Option (ii) are worlds where Ann did not pass but everyone else did. Here the weakly exhaustive answer to (154a) would be the proposition that Betty and Carl passed, the one to (154b) would be the proposition that Carl passed, and finally the one to (154c) would be the proposition that Betty passed. In other words, (152) is predicted to be utterable in case either Ann passed or she didn't pass and everyone else did. Crucially, (152) would be predicted to be grammatical.

In order to deal with this issue we suggest to slightly revise EXH^q along the lines in (155). In the case of *wh*-questions, instead of requiring that the weakly exhaustive answers to alternative *wh*-questions in w differ from the weakly exhaustive answer to the prejacent *wh*-question, (155) requires that the former asymmetrically entail the latter.

$$(155) \quad \llbracket EXH_{Alt_{str}}^q \rrbracket = \lambda p_{\langle s, st \rangle} . \lambda w_s . p(w)(w) = 1 \wedge \forall q \in Alt_{str} [p \not\Rightarrow q \rightarrow q(w) \Rightarrow p(w)]$$

This asymmetric entailment requirement cannot be satisfied given the K-extensions in (154). Since (154a) only partially overlaps with both of (155b) and (155c), it is impossible for the weakly exhaustive answer to both (155b) and (155c) to asymmetrically entail the one to (155a) at the same time. (152) thereby becomes unutterable.

While *but*-exceptives cannot felicitously occur in information-seeking *wh*-questions, they can appear in rhetorical questions that imply a negative answer 'no one', as noted by Hoeksema (1987). In these cases, the inference contributed by *but* is positive.

- (156) a. Who but God will help us now?
b. Who but a complete idiot would do this?

The advantage of our modification to the definition of EXH is that this fact follows naturally from the proposal. Assume the weakly exhaustive answer to the question in (156a) with God subtracted, i.e., *Who will help us now?*, is the tautology, i.e., its K-denotation is empty (recall footnote 26). We now reason about the alternatives formed by reintroducing God into the domain while subtracting other possible individuals. If God also cannot help, the alternatives will likewise be tautologies and, while they would entail the prejacent, they wouldn't asymmetrically do so. If God can help us, however, then the alternative weak answer will be the proposition that God can help us. This proposition does asymmetrically entail the weak answer of *Who will help us?*. The prediction is therefore that only in a world where no one can help us except for God the asymmetry between weak answers required by (155) is satisfied. That is, only in such worlds *wh*-questions with *but* will be utterable, yielding the rhetorical effect.

Thus, we successfully account for the difference between information-seeking *wh*-questions and rhetorical questions in their ability to host *but*. We predict that *wh*-phrases can only host *but* when the expected answer is *no one*, which is a desirable outcome.

7.4.2 Returning to *wh*-questions with *besides*

What are the consequences of the modification in (156) for *wh*-questions with *besides*? Consider (121a) and (121b) with plural marking on *student* again, once more repeated here:

- (157) a. Who besides Ann passed?
b. Which students besides Ann passed?

As discussed for the K-denotations in (146) in section 7.3.2, the only option for the weakly exhaustive answers to (157a) and (157b) to be non-equivalent to those of their respective alternatives without *besides* is for Ann to have passed. In these situations, it also holds that the weakly exhaustive answers to these alternative questions asymmetrically entail those to (157a) and (157b). The reason for this is that the K-denotations of the alternatives to (157a) and (157b) only differ from the K-denotations to the latter by having mention-some answers entailing Ann passing in them. That is, the weakly exhaustive answers can only be non-equivalent if those to the alternatives entail that Ann passed, which those to (157a) and (157b) can never entail.

This immediately raises the question of how to deal with the singular wh-question in (158). As discussed for the K-denotations in (142) in section 7.3.1 in order for both the uniqueness requirement of Ans_1 to be defined in the case of (158) and for its weakly exhaustive answer to differ from the one to the alternative without *besides*, it has to be the case that Ann plus exactly one other individual passed. But in that case the uniqueness requirement imposed by Ans_1 on the weakly exhaustive answer to the alternative is not satisfied. Consequently, it will not asymmetrically entail the weakly exhaustive answer to (158), as would be required by (156).

(158) Which student besides Ann passed?

The most straightforward way to deal with this issue is to further modify EXH^q by only requiring that a non-entailed weakly exhaustive answer to an alternative question asymmetrically entail the one to the prejacent question if the former has a defined value to begin with. This is done in (159).

$$(159) \quad \llbracket EXH_{Alt_{str}}^q \rrbracket = \lambda p_{\langle s, st \rangle} \cdot \lambda w_s \cdot p(w)(w) = 1 \wedge \forall q \in Alt_{str}[p \not\Rightarrow q \rightarrow [q(w)(w) \neq \# \rightarrow [q(w) \Rightarrow p(w)]]]$$

Consider the options for (158) in detail. In a world w where only Betty passed, uniqueness is satisfied for both weakly exhaustive answers but there will be no asymmetric entailment. w is thus not a possibility for (158). In a world w' where Betty and Carl passed, uniqueness is not satisfied for either of the weakly exhaustive answers. w' is also not a possibility for (159). Now, in world w'' where Ann and Betty passed uniqueness is satisfied for the weakly exhaustive answer to (158) but not for the one to its alternative. Since (159) does not impose asymmetric entailment from the latter to the former, w'' is the only option for (158). That is, the additive inference is derived.

7.4.3 Generalizing EXH

(159) can be generalized to be applicable to declarative prejacent. This can be brought about by modifying it so that the prejacent denotes a function taking only world arguments before finally returning a proposition, as in (160).

$$(160) \quad \llbracket EXH_{Alt}^p \rrbracket = \lambda p_{\langle s, \langle \dots, \langle s, t \rangle \rangle \rangle} \cdot \lambda w_s \cdot p(w) \dots (w) = 1 \wedge \forall q \in Alt[p \not\Rightarrow q \rightarrow [q(w) \dots (w) \neq \# \rightarrow [q(w) \Rightarrow p(w)]]]$$

When applied to a declarative prejacent, (160) is in effect equivalent to (161).

$$(161) \quad \llbracket EXH_{Alt_{str}}^p \rrbracket = \lambda p_{st} \cdot \lambda w_s \cdot p(w) = 1 \wedge \forall q \in Alt_{str}[p \not\Rightarrow q \rightarrow [q(w) \neq \# \rightarrow [q(w) \Rightarrow p(w)]]]$$

(161) requires that whenever a non-entailed alternative receives a defined truth-value in w , its truth-value in w must asymmetrically entail the one of the prejacent in w . Entailment between truth-values amounts to material implication:

$$(162) \quad \text{If } p, q \in D_t, p \text{ entails } q, p \Rightarrow q, \text{ iff } p = 0 \text{ or } q = 1.$$

Now, (161) also asserts the prejacent to be 1 in w . Given (162), asymmetric entailment can therefore only hold if the alternative is 0 in w . This means that (161) is actually equivalent to (163), which corresponds to the standard propositional *EXH*, modulo the definedness condition for $q(w)$ and the absent innocent exclusion condition. The latter could be easily added, as discussed in footnote 25.

$$(163) \quad \llbracket EXH_{Alt_{str}}^p \rrbracket = \lambda p_{st}. \lambda w_s. p(w) = 1 \wedge \forall q \in Alt_{str} [p \not\Rightarrow q \rightarrow [q(w) \neq \# \rightarrow q(w) = 0]]$$

In conclusion this means that the treatment of *besides* in *wh*-questions discussed above is fully compatible with exhaustification in the declarative domain and therefore also with our treatment of *besides* in declarative clauses.

8 Conclusion

In this paper, we proposed a unified compositional approach to the exceptive and additive uses of *besides*. To our knowledge, this is the first semantic account of *besides* that accurately captures its interaction with negative and universal quantifiers as well as all cases of modified numerals. We argued that the exceptive or additive inference arises from comparing the quantificational statement with domain subtraction performed by *besides* and the statement without this subtraction.

Our starting point was a unified treatment of *besides* with universal, negative, and non-monotonic quantifiers like *exactly n*. The resulting positive or negative inference depends on the properties of the quantifier: with universal quantifiers, the inference is predicted to be negative, whereas with negative and non-monotonic quantifiers, it is predicted to be positive. We extended this approach to all cases of modified numerals by adopting a decompositional account, assuming the presence of a constituent in the scope of the numeral equivalent to *exactly*. We demonstrated that the additive inference can be computed on this constituent. A key advantage of our proposal is its ability to unify the exceptive and additive uses of *besides* by appealing to well-established theoretical tools: focus marking, structural alternatives, and *EXH*.

We further extend our approach to *wh*-questions. Specifically, we suggest that *EXH* applies to the intension of the weakly exhaustive answer to the *wh*-question with *besides*. The set of alternatives for a *wh*-question with *besides* contains only the original question and the corresponding question without *besides*, consistent with the general assumptions in this paper. The K-extension of the original question is necessarily a proper subset of the K-extension of its only distinct alternative. We propose that *EXH* states that the extension of the alternative weakly exhaustive answer differs from that of the original question. This mechanism derives the additive inference.

Furthermore, we show that a minor modification of this idea, implemented in terms of asymmetric entailment, accounts for the observation that *but*-exceptives are compatible with *wh*-questions only in rhetorical contexts where the expected answer is negative. Finally, we generalize our definition of *EXH* so that it can apply uniformly to the intension of a weak answer to a question and to a proposition.

While this work covers more substantive number of contexts where *besides* is acceptable than any previous work, there remain a reach empirical ground for future research. We will briefly discuss some of these remaining issues.

In this work we have solely focused on the connected uses of *besides*, in other words on *besides* that follows the restrictor of the quantifier.

There is an interesting pattern involving preposed *besides* and existentials, as illustrated in (164b). In such cases, *besides* contributes an anti-containment inference—that is, the inference that the individual introduced by *besides* is not part of the restrictor set of the quantifier. This inference explains the unacceptability of (164b) (under the assumption that Ann is a girl), as contrasted with the perfect acceptability of (164c).

- (164) a. Some girls besides Ann passed the test.
 b. # Besides Ann, some girls passed the test.
 c. Besides John, some girls passed the test.

Preposed *besides* with universals does not exhibit this pattern: (165a) is acceptable and receives the familiar exceptive reading, including both the excepting inference and the containment inference that Ann is a girl. Interestingly, if the containment inference is not satisfied, a preposed *besides*, as in (165b), can only contribute the additive inference.

- (165) a. Besides Ann, every girl passed the test.
 b. Besides Bill, every girl passed the test.

While we do not have a proposal regarding the source of the anti-containment inference in (164b) or the derivation of the additive inference in (165b), we would like to point out that the containment inference is mandatory in the connected uses of *besides*, as captured by the proposal developed here.

- (166) a. #Some girls besides Bill passed the test.
 b. #Every girl besides Bill passed the test.

Another case we do not address here is preposed *besides* with focus associates, as in (167), which can only yield the additive meaning (Vostrikova 2019). Vostrikova (2019) suggests that the preposed instances without containment discussed above are, in fact, analogous to these cases, as *besides* does not operate on the domain of the quantifier but instead behaves similarly to (167c). This direction appears to be promising. In this case, the restriction observed in (164b) essentially is the restriction seen in (168) (unacceptable under the assumption that Ann is a girl).

- (167) a. Besides Ann, Bill_F danced with Mary.
 b. Besides Ann, Bill danced with Mary_F.
 c. Besides Ann, Bill_F passed the test.

- (168) #Ann came. Some girls came too.

Our theory potentially provides insight into why the additive reading with containment is not available for *besides* in (164b). According to our account, *besides* must be interpreted below the scope of *some*. If we assume that the surface position corresponds to the position where *besides* is interpreted, the mechanisms proposed here cannot apply in (164b).

Another interesting issue that we have to leave for future research is the interaction

between *besides* and quantifiers that are near the endpoints of the scale, such as *many* and *most*. *Besides* can felicitously occur with such quantifiers, as shown in (169). In these cases, *besides* is compatible with both the positive and negative inference about the individual introduced by *besides*. Evidence suggests that this is a genuine ambiguity. For instance, in (170), where only the cardinal interpretation of *many* is available, only the additive reading survives.

- (169) a. Most girls besides Ann passed the test.
 b. Many girls besides Ann passed the test.
- (170) There were many girls besides Ann at the gathering.
 $\rightsquigarrow$ Ann was there at the gathering

It is straightforward to see how the treatment proposed here can be extended to the cardinal reading of *many*. Indeed, we predict the additive inference in cases like (170), with the truth conditions paraphrased as follows:

- (171) There is a contextually large degree such that, if we exclude Ann, exactly this many people were at the gathering; but if we include her, the count is no longer exactly this number.

For *most* and *many* under the proportional reading, we predict a special kind of exceptive interpretation. Specifically, for (169), we predict a reading along the following lines: if we remove Ann from the restrictor, it is true that the majority of girls passed the test; however, if we consider the whole domain, it is false that the majority of girls passed. This reading captures the exceptive inference that Ann did not pass, but it also requires that the inclusion or exclusion of Ann makes a crucial difference to the validity of the *most*-claim. Alas, our informants tell us that the sentence does not require this to be the case. Thus, our current proposal does not provide a solution for how the exceptive reading arises for *most* and *many* or how the additive reading with *most* is derived.

Native speakers of English reject the corresponding examples with *but* as ungrammatical.

- (172) a. *Most girls but Ann passed the test.
 b. *Many girls but Ann passed the test.

These examples contrast with the perfectly acceptable cases reported in Peters and Westersthål (2023), where *but* is compatible with *most* and *many*, contributing a negative inference in such contexts, like the one in (173). Again, the reported judgment for such cases is that including Sundays does not necessarily make the *most*-claim false.

- (173) We are available most days but Sundays.

We remain optimistic that a compositional theory in terms of *EXH* incorporating the key insights from Peters and Westersthål (2023) can be developed to account for all the contrasts observed here. For now, we must leave this to future research.

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